e-RoomReserve: AI-Driven Smart Campus Space Optimization System with Real-Time Occupancy Detection and Autonomous Reservation Control using Bluetooth Low Energy (BLE) Technology

A Capstone Project

Presented to the Faculty of the

School of Communication, Multimedia, and Computer Studies

Information Technology Department

St. Dominic College of Asia

City of Bacoor, Cavite

Submitted by:

Agoncillo, John Cyrus

Echepari, Adrian V.

Magno, Joseph Emmanuel A.

Otchia, Cybeir Kim O.

May, 2026

Acknowledgements

The researchers would like to express their sincere gratitude to the following individuals and institutions whose guidance, support, and encouragement made the completion of this capstone project possible.

To Dr. Alvin Jason Virata, their capstone adviser, for his invaluable mentorship, constructive feedback, and unwavering support throughout the duration of this project.

To the Department of Information Technology of St. Dominic College of Asia, for providing the academic environment and resources necessary for the development of this system.

To the faculty, staff, and students of St. Dominic College of Asia who participated in interviews, surveys, and user acceptance testing, for generously sharing their time and insights.

To their families and friends, for their patience, encouragement, and moral support.

Above all, to God Almighty, for the wisdom, strength, and perseverance to complete this undertaking.

Abstract

e-RoomReserve is an AI-integrated room reservation and occupancy monitoring system developed for St. Dominic College of Asia. The system addresses inefficiencies in the institution's manual room scheduling process by combining a mobile and web application, Bluetooth Low Energy (BLE) beacon-based presence detection using the ESP32 module, a faculty approval workflow, automated staff notifications, an AI-based room recommendation module using content-based filtering, and a sentiment analysis component powered by VADER. The mobile application enables students and faculty to submit reservation requests and perform BLE-verified check-ins, while the web-based administrative portal provides real-time occupancy monitoring, approval management, and usage analytics. The system's architecture follows a four-layer model comprising a Presentation Layer, Application Layer, Data Layer powered by Firebase and Supabase, and a Device Layer consisting of ESP32 BLE beacons. Development follows an Agile methodology using the Scrum framework across seven sprints. Evaluation is conducted through unit testing, integration testing, and user acceptance testing. e-RoomReserve aims to improve space utilization, reduce administrative burden, and provide a scalable foundation for future smart campus initiatives at St. Dominic College of Asia.

Keywords: room reservation system, occupancy monitoring, Bluetooth Low Energy, BLE beacon, ESP32, AI recommendation, content-based filtering, sentiment analysis, VADER, smart campus, IoT, Firebase

St. Dominic Complex, E. Aguinaldo Highway, Bacoor City, Cavite, Philippines 4102
0917 625 8222 • 0922 850 9200 • (046) 417 7322 • (046) 417 8161
www.stdominiccollege.edu.ph stdominiccollege @sdcapikemen
SCHOOL OF COMMUNICATION, MULTIMEDIA & COMPUTER STUDIES

Table of Contents

Chapter I Introduction…….

1.1 Background of the Study……... 1

1.2 Statement of the Problem……..

1.3 Objectives of the Study…….

1.4 Scope and Limitations of the Study……..

1.5 Concept Framework……..……

1.6 Definition of Terms……...

1.7 Significance of the Study……..……

Chapter II Review of Related Literature and Studies……...

2.1 Internet of Things in Smart Building and Facility Management……...

2.2 AI-Based Recommendation System and Content-Based Filtering……...

2.3 Bluetooth Low Energy (BLE) for Indoor Presence Detection……..

2.4 Digital Room Reservation System and Space Utilization Monitoring……..

2.5 Smart Energy Management Through Occupancy Monitoring……..

2.6 System Architecture and Security in IoT Monitoring System……..

2.7 Smart Campus Monitoring in the Philippine Context……...

2.8 Research Gap…….

2.9 Synthesis of Related Literature…….

Chapter III Project Design and Development……

3.1 System Objective……...

3.2 System Framework……

3.3 Feasibility Study……

3.3.1 Technical Feasibility……

3.3.2 Operational Feasibility……

3.3.3 Economic Feasibility……...

3.4 System Design - SDLC…….

3.4.1 User Interface……..

3.4.2 System Input and Output…….

3.5 Gantt Chart……

3.6 Testing Process……..

3.6.1 Unit and System Testing……..

3.6.2 Integration Testing……...

3.6.3 User Acceptance Testing (UAT).....

3.6.4 Performance Benchmarks……

3.7 Installation and Deployment…….

3.8 Maintenance……..

References……..

Chapter I

Introduction

St. Dominic College of Asia manages multiple classrooms and meeting spaces that require efficient scheduling and monitoring to maximize utilization. Traditional reservation systems rely on manual booking and periodic physical checks, which lead to no-shows, double bookings, unverified use of spaces, and inefficient space allocation. Advances in Internet of Things (IoT) technologies — specifically Bluetooth Low Energy (BLE) beacon technology — enable low-cost, real-time verification of room occupancy. This project proposes a campus-tailored solution that integrates a mobile and web reservation application, a faculty approval workflow, BLE-based presence detection and occupancy monitoring, an automated staff notification system, and an AI-based room recommendation feature that assists users in selecting appropriate spaces based on availability and room attributes, while incorporating post-reservation feedback analysis to support facility management decisions.

1.1 Background of the Study

Room scheduling and space management are fundamental concerns in educational institutions, where classrooms, laboratories, and meeting rooms must be allocated efficiently among faculty, students, and administrative personnel. At St. Dominic College of Asia, room reservations are currently managed through a manual process that relies on digital logbooks, verbal coordination with designated personnel, and periodic visual inspection to confirm whether a reserved space is actually in use. While this approach has served the institution for many years, it has become increasingly inadequate as enrollment grows, school organizations develop, and demand for shared spaces rises.

The existing reservation procedure requires users to submit requests either in person or through informal communication channels, after which a faculty member or administrator manually records and validates the booking. There is no centralized digital system to track reservations in real time, verify occupancy, or automatically release rooms that go unused after a scheduled period. As a result, double bookings, miscommunications, and unverified use of spaces

occur regularly, disrupting academic and organizational activities and creating friction between departments and user groups.

Beyond scheduling inefficiencies, the absence of automated occupancy monitoring creates significant gaps in real-time visibility over room usage. Manual reservation systems provide no mechanism for detecting early vacancies, no-shows, or overtime occupancy — situations that occur outside the predictable schedule that staff rely on for routine room checks. As a result, rooms may sit unused for extended periods while remaining marked as reserved, blocking other users from accessing available spaces. Without real-time occupancy data, administrators also lack the usage records needed to identify recurring patterns such as frequent no-shows or overtime occupancy — information that is essential for enforcing reservation policies and optimizing space allocation decisions (German et al., 2021; Somwong et al., 2022). By integrating occupancy detection and automated notifications, e-RoomReserve addresses these gaps through real-time BLE-based presence monitoring and structured staff alert mechanisms.

In addition to scheduling conflicts, users often struggle to identify suitable rooms that match their capacity requirements or availability windows, which further contributes to inefficient space utilization. Intelligent recommendation features in digital reservation systems can assist users in selecting appropriate spaces based on real-time availability and historical usage patterns.

Several academic and commercial institutions in the Philippines and abroad have begun adopting IoT-based facility management solutions. Smart campus initiatives have explored sensor-based monitoring to improve campus operations and resource management (Taruc & De La Cruz, 2024), and the Department of Science and Technology has promoted IoT integration in educational systems as part of broader digital transformation initiatives (DOST, 2025).

This study responds to that gap by designing and developing an AI-Integrated Room Reservation and Occupancy Monitoring System specifically tailored to the operational context of St. Dominic College of Asia. The proposed system combines a mobile application and web application accessible to students, faculty, utility staff, and a building administrator interface for facility management, BLE beacon presence detection, and an AI-based room recommendation feature that assists users in identifying appropriate rooms based on availability and room attributes. These components are connected through a cloud backend powered by Firebase and a Next.js server layer. The system also records usage data and feedback that can be analyzed to

support scheduling optimization and facility management decisions. The goal is to modernize room management at the institution, improve the accuracy of occupancy data, and lay the groundwork for broader smart campus initiatives.

1.2 Statement of the Problem

St. Dominic College of Asia faces several operational challenges related to room reservations and occupancy, including inefficient room utilization caused by unverified bookings, frequent no-shows, and extended use of rooms beyond scheduled times that lead to conflicts. The institution also experiences a significant administrative burden on faculty and staff who must manually approve, monitor, and enforce reservations. Users also lack a system that can assist them in selecting the most suitable rooms based on availability and room attributes such as capacity, type, and amenities, which can lead to inefficient space allocation. Additionally, the lack of real-time occupancy data limits effective planning, reporting, and maintenance decisions. To address these issues, this study proposes the design and development of an AI-integrated room reservation and occupancy monitoring system that provides reliable occupancy verification through BLE-based detection, automated vacancy detection, structured approval workflows, and an AI-based room recommendation feature to assist users in selecting suitable spaces based on availability and room attributes.

1.3 Objectives of the Study

The study aims to design and develop an AI-integrated room reservation and occupancy monitoring system using Bluetooth Low Energy (BLE) beacon technology to improve space utilization, enforce reservation policies, and support efficient facility management at St. Dominic College of Asia. Specifically, it aims to:

1. Design and develop a mobile and web application that support paperless room reservation requests, faculty approval workflows, reservation status tracking, and user notifications for students, faculty, and utility staff of St. Dominic College of Asia, with additional interfaces for building administrators to manage facilities and super administrators to manage system-wide user roles.
2. Develop a BLE beacon–based presence detection mechanism using the ESP32 BLE Beacon Module that requires users to be physically near the room beacon before

activating a reservation, monitors room occupancy in real time, and automatically marks rooms as vacant when the registered user's device leaves beacon range after a configurable timeout period.

3. Develop an automated alert system that notifies staff when BLE signal loss is detected during an active reservation, when rooms are occupied past scheduled end times, and when physical room inspection is required as a fallback verification measure.
4. Design and implement a structured reservation workflow that reduces scheduling conflicts, unverified room usage, and administrative workload through faculty-verified check-in, role-based access control, and automated conflict detection.
5. Develop an AI-based room recommendation module that suggests suitable alternative rooms based on attributes such as capacity, amenities, and size when a user's preferred room is unavailable, and incorporates sentiment analysis of post-reservation feedback from users and staff to support facility maintenance decisions.
6. Analyze room usage logs, occupancy data, and post-reservation survey responses to generate actionable insights for faculty and administrators, supporting scheduling optimization, policy enforcement, and facility management decisions.
7. Evaluate the system's performance, usability, and effectiveness through unit testing, integration testing, and user acceptance testing to determine the extent to which e-RoomReserve meets its operational objectives in the context of St. Dominic College of Asia.

1.4 Scope and Limitations of the Study

1.4.1 Scope

- Design and prototype implementation of e-RoomReserve, an AI-integrated room reservation and occupancy monitoring system for selected classrooms at St. Dominic College of Asia.
- Mobile and web applications accessible to students, faculty professors, and utility staff with features such as room reservation requests, BLE beacon-based proximity verification for reservation activation, real-time occupancy monitoring, reservation status tracking, push notifications, AI-based alternative room recommendations,

post-reservation survey submission, and reservation approval management for faculty professors.

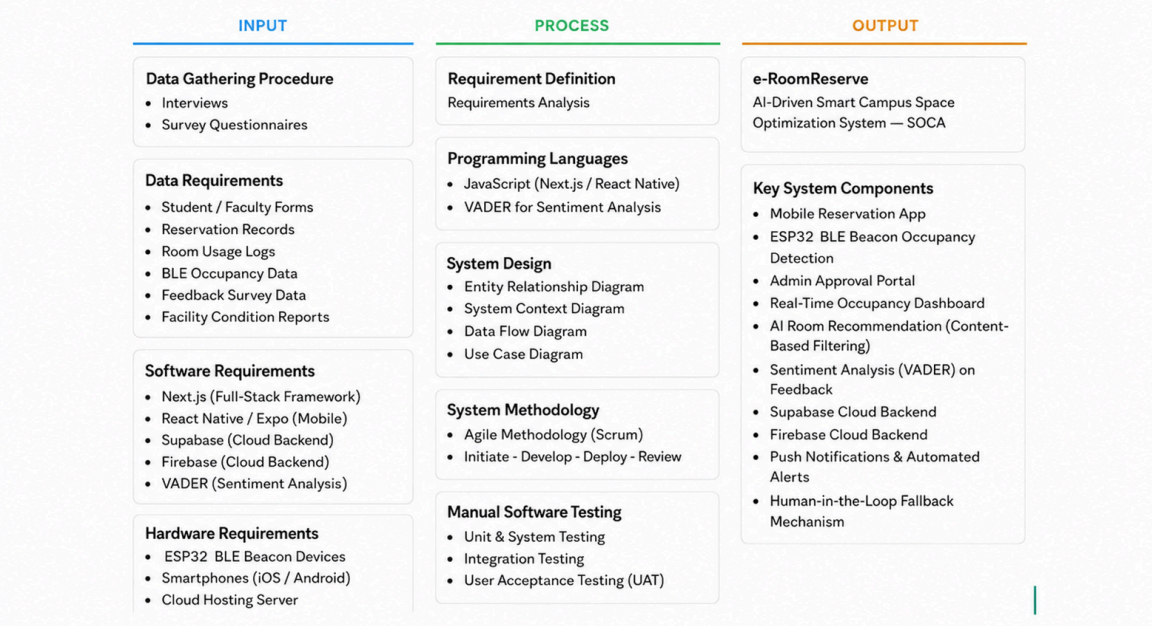
- Web application with role-specific interfaces for building administrators to manage reservations, configure rooms, floors, and buildings, and super administrators to manage system-wide user roles and account approvals.
- ESP32 BLE beacon–based presence detection mechanism that monitors room occupancy through BLE client connection state, automatically triggering staff alerts and prompting physical verification before a room is marked as vacant.
- AI-based room recommendation module that assists both students and faculty in selecting appropriate rooms based on availability, capacity, and room attributes such as room type, soundproofing, amenities, and size.
- Post-reservation survey system that collects feedback from both students and staff upon completion of a reservation, with sentiment analysis applied to free-text responses to generate insights on room condition, user satisfaction, and facility maintenance needs.
- Hardware components consist of ESP32 BLE Beacon Module for presence detection, USB power adapters for continuous beacon power supply, and protective enclosure boxes for secure beacon installation, all supported by a cloud-based backend for data storage and processing.
- Human-in-the-loop fallback mechanism wherein automated BLE occupancy detection is supplemented by staff physical verification, ensuring that room status changes are confirmed by human judgment rather than signal data alone.

1.4.2 Limitations

- BLE-based proximity verification using RSSI is used to ensure users are physically near the room before activating a reservation. RSSI values may be affected by environmental factors such as walls, furniture, and interference, potentially preventing reservation activation even when the user is near the room or allowing activation from just outside it.
- Unverified room use may still occur outside the system if users occupy spaces without submitting a reservation request, as the system has no mechanism for detecting occupancy by individuals who have not initiated a reservation through e-RoomReserve.
- Occupancy detection features depend on mobile device compatibility and reliable network connectivity, and outages could temporarily reduce functionality.

- Continuous Bluetooth activation cannot be enforced programmatically; the system mitigates this by automatically alerting staff when signal loss is detected during an active reservation, enabling manual verification and user reminders as a fallback measure.
- The system is unable to detect the number of occupants in a room, as BLE-based occupancy detection is limited to monitoring the connection state of the registered reservation holder's mobile device and cannot identify or count additional individuals occupying the space.
- Privacy concerns related to presence detection require careful handling of personal device identifiers, explicit user consent, and measures to anonymize stored data.
- The sentiment analysis module relies on free-text survey responses from users and staff, and its accuracy may be affected by incomplete, very short, or ambiguous responses. Insights generated from insufficient feedback data may not accurately reflect actual room conditions or user satisfaction.
- Prototype evaluation will be limited to a subset of rooms, and campus-wide deployment will require additional testing, user training, and resource planning.

1.5 Conceptual Framework



The development of e-RoomReserve is guided by a structured conceptual framework organized around three primary phases: Input, Process, and Output, culminating in an Evaluation stage.

The Input phase encompasses the data, software, and hardware requirements necessary to initiate the development of the system. Data gathering is conducted through interviews and survey questionnaires, producing information on student and faculty forms, reservation records, room usage logs, BLE occupancy data, feedback survey data, and facility condition reports. On the software side, the system requires Next.js as the full-stack web framework, React Native with Expo for the mobile application, Firebase as the cloud backend, Supabase for file storage and upload management — chosen for its cost-effectiveness as a free alternative to Firebase Storage — and VADER for sentiment analysis. Hardware requirements consist of ESP32 BLE Beacon devices, USB power adapters, protective enclosure boxes, smartphones running iOS or Android, and a cloud hosting server.

The Process phase covers the steps undertaken to design, develop, and test the system. It begins with requirement definition through requirement analysis, followed by the selection of

programming languages — specifically JavaScript via Next.js and React Native for both the web and mobile components, including the sentiment analysis module which is implemented using the VADER library through the JavaScript ecosystem. System design involves the creation of an Entity-Relationship Diagram, System Context Diagram, Data Flow Diagram, and Use Case Diagram to model the system's structure and behavior. Development follows an Agile Methodology using the Scrum framework, consisting of Initiate, Develop, and Deploy-Revise cycles. The process concludes with manual software testing across three levels — Unit and System Testing, Integration Testing, and User Acceptance Testing.

Lastly, the output phase presents e-RoomReserve as the primary deliverable — an AI-Integrated Room Reservation and Occupancy Monitoring System developed for St. Dominic College of Asia. The system's key components include a Mobile Reservation App, BLE Beacon Occupancy Detection using the ESP32 BLE Beacon Module, an Admin Approval Portal, a Real-Time Occupancy Dashboard, an AI Room Recommendation module using Content-Based Filtering, Sentiment Analysis on feedback via VADER, a Firebase Cloud Backend, Supabase Storage for file and image uploads, selected as a free alternative to Firebase Storage — Push Notifications and Automated Alerts, and a Human-in-the-Loop Fallback Mechanism.

1.6. Definition of Terms

- ***Anonymization*** **-** A set of techniques used to remove or obscure personal identifiers from stored data to protect user privacy and comply with data protection principles.
- ***API (Application Programming Interface)*** **-** A collection of web-based endpoints that enable communication between the mobile application, administrative portal, and backend services, ensuring seamless system integration.
- ***Artificial Intelligence (AI)*** **-** A branch of computer science concerned with building systems capable of performing tasks that typically require human intelligence, such as pattern recognition, decision-making, and prediction.
- ***Authentication*** **-** The process of verifying a user's identity, typically through login credentials, to ensure secure access to the system.
- ***Authorization*** **-** The mechanism that determines and grants user permissions based on authenticated identity and assigned roles, such as student, faculty, or administrator.

- ***Content-Based Filtering -*** A recommendation technique that suggests items to a user based on the attributes or features of items they have previously selected or shown interest in, rather than on the behavior of other users. It compares item characteristics to identify the most similar available alternatives.
- ***Firebase -*** A cloud-based platform developed by Google that provides real-time database, authentication, cloud storage, and serverless backend services.
- ***ESP32 BLE Beacon Module -*** A low-cost, low-power system-on-chip microcontroller developed by Espressif Systems that integrates Bluetooth Low Energy (BLE) and Wi-Fi connectivity, making it ideal for IoT applications.
- ***Human-in-the-Loop -*** A system design approach in which human judgment and intervention are incorporated as a deliberate component of an otherwise automated process.
- ***HVAC (Heating, Ventilation, and Air Conditioning) -*** Systems designed to regulate indoor temperature, humidity, and air quality in buildings. They consist of various components like furnaces, air conditioners, and ventilation systems to ensure comfort and maintain healthy air quality.
- ***iBeacon -*** A Bluetooth Low Energy broadcasting protocol developed by Apple that allows devices to transmit a universally unique identifier (UUID) along with Major and Minor values.
- ***Internet of Things (IoT) -*** A network of interconnected physical devices embedded with sensors, software, and communication capabilities that collect and exchange data over the internet without requiring direct human interaction.
- ***Next.js -*** A full-stack web development framework built on React that supports both server-side rendering and static site generation.
- ***Occupancy Detection -*** The process of determining whether a physical space is currently in use by monitoring signals or sensor data.
- ***Push Notification -*** A message automatically sent from a server to a user's mobile device or desktop interface without the user actively requesting it.
- ***React Native / Expo -*** A cross-platform mobile application development framework that allows developers to build iOS and Android applications using JavaScript and React.

Expo extends React Native with additional tools and libraries for simplified development and deployment.

- ***Recommendation System -*** An algorithm-driven module that analyzes user data, historical patterns, and contextual factors to suggest relevant options to a user.
- ***Reservation Engine -*** The backend system component responsible for managing booking requests, approval workflows, scheduling rules, and conflict resolution to ensure proper room allocation.
- ***Room Reservation System -*** A digital platform that enables users to request, schedule, and manage the use of physical spaces within an institution.
- ***RSSI (Received Signal Strength Indicator) -*** A measurement of signal strength received from a BLE beacon. RSSI values are used to approximate the distance between a mobile device and the beacon for occupancy detection.
- ***Sentiment Analysis (SA) -*** A natural language processing technique used to automatically identify and classify the emotional tone expressed in text as positive, negative, or neutral. It is commonly applied to user-generated content such as reviews, survey responses, and comments to extract meaningful insights from unstructured textual data.
- ***Smart Building -*** A structure equipped with interconnected IoT sensors, automated control systems, and data analytics capabilities that enable real-time monitoring and management of building operations such as lighting, HVAC, security, and space utilization.
- ***Supabase -*** An open-source backend-as-a-service platform that provides file storage, a relational PostgreSQL database, authentication, and RESTful APIs.
- ***Timeout Logic -*** A configurable time interval used to determine when a room should be marked vacant after a user's device is no longer detected within BLE range.
- ***VADER (Valence Aware Dictionary and sEntiment Reasoner) -*** A free, open-source, rule-based sentiment analysis tool designed to evaluate the sentiment of short, informal texts. It uses a validated lexicon of sentiment-bearing words combined with grammatical rules to produce sentiment scores without requiring model training or external API services.

1.7 Significance of the Study

Before outlining the specific beneficiaries, it is important to emphasize that this study supports the modernization of facility management within St. Dominic College of Asia. By developing an AI-integrated reservation and occupancy monitoring system, the research enhances efficiency, transparency, and data-driven decision-making in classroom utilization. The system reduces manual processes, minimizes scheduling conflicts, and provides real-time insights that support better resource management and campus operations. This study is beneficial to the following stakeholders:

Students. Students will have access to a convenient and user-friendly digital platform for reserving rooms and viewing real-time availability. This reduces uncertainty, prevents double bookings, and saves time previously spent coordinating manually. The ability to receive notifications and verify reservations through secure check-in methods also ensures a smoother and more reliable experience when using campus facilities.

Faculty Members. Faculty members will benefit from a more organized and structured system for managing room reservations. The platform streamlines the process of requesting and confirming room use, minimizing last-minute conflicts and unverified occupancy. With clear reservation records and automated monitoring, faculty can focus more on instruction and academic responsibilities rather than administrative coordination.

School Administration. The administration will gain improved monitoring, reporting, and decision-making capabilities regarding facility utilization. Access to occupancy dashboards and analytical reports allows administrators to identify trends, optimize room allocation, and make informed decisions about scheduling, maintenance, and facility management. This contributes to better resource planning and institutional efficiency.

Information Technology Department. The system may serve as a foundation for future smart campus initiatives and further Internet of Things (IoT) integration within the institution. By establishing a scalable architecture that combines mobile applications, hardware sensors, and cloud-based services, the IT department can expand the platform to support additional building automation features and data-driven campus management solutions.

Future Researchers. The study may serve as a reference for future research related to IoT-based reservation systems and smart campus technologies. It provides a practical framework,

implementation insights, and evaluation findings that can inform subsequent studies seeking to enhance AI-integrated facility management systems in educational institutions.

Chapter II

Review of Related Literature and Studies

This chapter presents a review of related literature and studies that inform the design and development of e-RoomReserve, an AI-integrated room reservation and occupancy monitoring system integrating Bluetooth Low Energy (BLE) beacon technology. The discussion covers key areas relevant to the proposed system, including smart facility and space management, BLE-based indoor presence detection, mobile reservation platforms, IoT system architecture, cloud-based data management, and AI-based recommendation systems. The review draws from recent academic studies, institutional implementations, and industry developments to establish a theoretical and empirical foundation for the system's core components and to situate e-RoomReserve within the broader landscape of smart campus technology.

2.1 Internet of Things in Smart Building and Facility Management

The Internet of Things (IoT) has emerged as a transformative force in the management of buildings and facilities, enabling real-time monitoring, automation, and data-driven decision-making across a wide range of built environments. At its core, IoT refers to a network of interconnected physical devices embedded with sensors, software, and communication capabilities that collect and exchange data without direct human intervention. In the context of smart buildings, these capabilities have been applied to improve occupant comfort, enhance security, optimize space utilization, and streamline facility operations.

Chaudhari et al. (2024) provide a comprehensive overview of IoT-based occupancy detection for smart buildings, noting that occupancy information is central to automating building functions and improving operational efficiency. Their review establishes that a typical IoT architecture for smart buildings involves sensors collecting environmental and occupancy data, which are then transmitted to a data fusion center, processed through analytics algorithms, and used to issue commands to a building control system. This layered architecture — where sensing, processing, and control functions are clearly separated yet interconnected — closely mirrors the design approach of e-RoomReserve, which similarly relies on ESP32 BLE beacon modules as the

sensing layer, a Next.js backend for data processing, and Firebase for cloud-based data management and synchronization.

Building on this foundation, Khan et al. (2024) examine the specific role of IoT technologies in occupancy prediction, arguing that real-time occupancy awareness enables buildings to operate more efficiently by ensuring that facilities and services are available only when and where they are needed. Their review emphasizes that smart and cognitive buildings — those augmented with sensing, actuation, and data processing capabilities — can dynamically respond to occupancy conditions, improving both resource utilization and the overall operational experience for building occupants. Importantly, Khan et al. (2024) also identify Bluetooth Low Energy (BLE) as one of several key non-mobility sensing technologies used for occupancy detection, noting that BLE-enabled devices can function as proximity beacons to enable accurate, real-time monitoring of room occupancy — directly relevant to the detection approach employed in e-RoomReserve.

The application of IoT for occupancy monitoring in university buildings specifically is examined by Franco et al. (2025), who deployed a low-cost IoT sensor network across multiple classroom buildings at the University of Pisa. Their study demonstrated that integrating real-time occupancy data with building management systems enabled more responsive and efficient facility operations, with Franco et al. (2025) noting that university buildings present unique management challenges due to their large size, fluctuating occupancy patterns, and the diversity of spaces they contain. These characteristics make real-time occupancy data particularly valuable for operational decision-making — findings that directly support the rationale for e-RoomReserve's occupancy monitoring component, which aims to improve space utilization and reduce unnecessary facility usage by automatically detecting the presence or absence of users in reserved rooms.

At a broader level, Olimat et al. (2023) conducted a bibliometric analysis of smart facility management technologies, examining 7,236 publications from 2012 to 2023. Their analysis reveals that IoT is the second most researched technology in facility management after Building Information Modeling (BIM), with strong research connections to smart building systems and real-time asset monitoring. The study notes that IoT's integration with cloud platforms, artificial intelligence, and sensor networks has established the conceptual and technical groundwork for

AI-integrated facility management systems — including automated occupancy monitoring and reservation management — that were not feasible in earlier manual approaches.

Collectively, these studies establish a well-supported theoretical and empirical basis for the IoT-driven approach adopted by e-RoomReserve. The convergence of real-time occupancy sensing, cloud-based data management, and structured monitoring logic — as demonstrated across smart building research — directly informs the system's architecture, particularly its BLE beacon-based presence detection, cloud-based data management, and administrative monitoring capabilities.

2.2 AI-Based Recommendation Systems and Content-Based Filtering

Recommender systems have become a fundamental component of modern software platforms, enabling automated, data-driven suggestions that assist users in navigating large sets of available options. Ricci et al. (2022) define recommender systems as software tools and techniques that provide suggestions for items that are most likely of interest to a particular user, covering approaches such as content-based filtering, collaborative filtering, and hybrid methods. In the context of e-RoomReserve, content-based filtering is the most applicable approach — when a user attempts to reserve a room that is already occupied or reserved, the system automatically identifies and suggests alternative rooms whose attributes most closely match those of the originally selected room, filtered by availability for the requested time and date.

Content-based filtering operates by comparing the properties of items rather than the behavior of other users, making it well-suited for structured domains where items have clearly defined attributes. In e-RoomReserve, each room is characterized by attributes including seating capacity, room type, soundproofing, air conditioning status, available equipment, number of electrical outlets, and floor area. When a room is unavailable, the system queries these attributes across all available rooms and ranks alternatives by similarity to the original selection. This approach requires no historical user data to function, making it particularly appropriate for a prototype deployment where usage history may be limited.

Beyond attribute-based recommendation, e-RoomReserve incorporates a feedback-driven layer that refines recommendation quality over time. Upon completion of each reservation, both the student and the assigned staff member are prompted to complete post-use surveys. Student responses capture satisfaction with the room and the overall system experience, while staff

responses record the physical condition of the room after use, including cleanliness and missing equipment. Free-text responses from both surveys are processed using VADER (Valence Aware Dictionary and sEntiment Reasoner), a lexicon-based sentiment analysis tool well-suited for short, informal user-generated text. Syazwani et al. (2020) applied VADER to student feedback responses in an e-learning environment, finding that it achieved approximately 90% accuracy in sentiment classification, outperforming both Naïve Bayes and Decision Tree machine learning classifiers, supporting its selection for processing post-reservation survey responses in e-RoomReserve. The analyzed feedback is aggregated into per-room sentiment scores and condition flags, which are presented to faculty and administrators through the system dashboard as auto-generated insights — such as identifying rooms with consistently poor thermal comfort or flagging users who repeatedly leave rooms in poor condition. These insights support facility maintenance decisions while also feeding back into the recommendation engine to deprioritize rooms with poor feedback histories in future suggestions.

2.3 Bluetooth Low Energy (BLE) for Indoor Presence Detection

Bluetooth Low Energy (BLE) has emerged as one of the most widely adopted technologies for indoor presence detection and occupancy monitoring, owing to its low power consumption, low cost, and near-universal compatibility with modern smartphones. BLE beacons operate by continuously broadcasting advertisement packets at configurable intervals, which are detected by nearby mobile devices and used to infer proximity through the Received Signal Strength Indicator (RSSI) — a measurement of the signal strength received from the beacon that decreases as distance increases.

Demrozi et al. (2021) investigated the use of low-cost BLE devices for estimating indoor occupancy in public buildings, demonstrating that BLE-based presence detection provides a practical and cost-effective mechanism for determining whether a space is occupied, without requiring complex sensor infrastructure. Their study confirms that BLE beacons — when paired with a mobile application continuously scanning for beacon signals — enable reliable room-level occupancy determination that can be used to support automated facility management decisions. This architecture directly mirrors the design of e-RoomReserve, where an ESP32 BLE beacon

module is mounted in each room and the mobile application monitors signal presence to determine occupancy status in real time.

The ESP32 microcontroller has been specifically evaluated as a BLE-based indoor positioning device. Khanh et al. (2021) implemented a BLE-based indoor positioning system using the ESP32-NodeMCU, demonstrating that the ESP32 is a practical and deployable hardware platform for BLE-based indoor detection applications. While e-RoomReserve does not require precise coordinate-level positioning — only room-level presence confirmation — this study validates the ESP32 as a functional BLE beacon hardware choice and highlights the environmental factors that must be accounted for during installation and calibration.

The accuracy limitations of RSSI-based detection are further examined by Janczak et al. (2022), who conducted a detailed accuracy analysis of a BLE indoor location system based on RSSI measurements across varying distances and environmental conditions. Their findings demonstrated that BLE RSSI provides reliable proximity estimation within close range under line-of-sight conditions, but that signal variability increases significantly in obstructed environments — consistent with the classroom setting of e-RoomReserve where walls, furniture, and room occupants can introduce interference. In e-RoomReserve, RSSI is used specifically for proximity verification during reservation activation, ensuring users are physically near the room before initiating a reservation. Occupancy monitoring throughout the reservation period relies on BLE client connection state rather than continuous RSSI estimation, with staff physically verifying room status before any vacancy changes are confirmed — mitigating the impact of temporary signal fluctuations on occupancy records.

Collectively, these studies establish BLE beacon technology as a well-validated, low-cost, and sufficiently accurate approach for the room-level presence detection requirements of e-RoomReserve, while grounding the system's known RSSI-related limitations in established academic literature.

2.4 Digital Room Reservation Systems and Space Utilization Monitoring

The management of shared spaces in educational and institutional settings has increasingly shifted toward digital reservation platforms that combine scheduling functionality

with real-time occupancy data. Traditional manual booking systems — reliant on logbooks, verbal coordination, and periodic physical checks — are widely recognized as insufficient for environments with high and variable demand for shared spaces, as they cannot verify whether reserved rooms are actually in use or automatically release spaces that go unused.

Zakarija et al. (2025) present an overview of an online room booking system developed for university settings, addressing the inefficiency of manual timetable creation and the problems caused by unsynchronized scheduling across departments. Their system focuses on the data, business, and presentation layers of a reservation platform, with particular attention to the administrative processes of higher education institutions — including the structured management of room requests, approvals, and real-time availability checks. The study highlights that providing students and staff with an intuitive platform for reserving rooms for different types of activities, combined with real-time availability checking, enables better coordination and resource optimization. This institutional framing closely mirrors the operational context of e-RoomReserve, which similarly targets the scheduling inefficiencies and administrative burden associated with manual room management at St. Dominic College of Asia.

The importance of integrating occupancy detection directly into reservation workflows is demonstrated by Somwong et al. (2022), who proposed a smart room vacancy and booking system that uses IoT sensors collecting temperature, sound, and light intensity data to train a machine learning model for real-time vacancy prediction. Their system pairs this automated occupancy detection with a web-based reservation interface, allowing room availability status to reflect actual physical occupancy rather than scheduled bookings alone. This directly addresses one of the most persistent problems in manual reservation systems — the gap between scheduled and actual room usage — which e-RoomReserve similarly bridges through its ESP32 BLE beacon-based presence detection mechanism. Somwong et al. (2022) reported very high accuracy and F1-scores in room vacancy prediction, demonstrating that sensor-based occupancy verification is a technically viable and reliable approach for smart reservation systems.

Together, these studies establish that effective digital room reservation systems must go beyond simple scheduling to incorporate real-time occupancy verification, automated status updates, and structured administrative workflows — precisely the capabilities that

e-RoomReserve seeks to deliver through the integration of BLE-based presence detection, a faculty approval portal, and an occupancy monitoring dashboard.

2.5 Smart Energy Management Through Occupancy Monitoring

The relationship between real-time occupancy monitoring and facility management efficiency has been widely studied, with consistent findings that automated occupancy detection significantly improves space utilization, reduces unnecessary facility usage, and supports more informed operational decision-making. In educational institutions particularly, where rooms are frequently left occupied past their reserved times or remain marked as reserved despite being vacant, real-time occupancy data provides administrators and staff with the visibility needed to manage spaces more effectively.

Chaudhari et al. (2024) establish that occupancy detection is central to the operational management of building systems, noting that knowing whether a space is occupied at any given time enables institutions to make timely and accurate decisions about facility usage, maintenance scheduling, and resource allocation. Their review further highlights that IoT-based occupancy monitoring reduces the reliance on manual inspections and periodic physical checks, freeing staff to focus on higher-priority tasks rather than routine room monitoring. These findings provide direct academic grounding for e-RoomReserve's occupancy monitoring component, which automatically notifies staff when BLE signal loss is detected during an active reservation — enabling timely physical verification and informed decision-making about room status without requiring constant manual oversight.

The specific application of IoT-based occupancy monitoring in university buildings is examined by Franco et al. (2025), who deployed a low-cost IoT sensor network across multiple classroom buildings at the University of Pisa. Their study demonstrated that real-time occupancy data enables building managers to respond more promptly to changes in room usage, reducing the gap between actual and recorded occupancy states. Franco et al. (2025) further noted that university buildings present unique facility management challenges due to fluctuating occupancy patterns and the tendency for rooms to remain in use — or appear in use — beyond their scheduled periods, creating inefficiencies in space allocation and administrative coordination. These challenges closely mirror the operational context of St. Dominic College of Asia described

in Chapter 1, where the absence of real-time occupancy data prevents staff from responding promptly to unused or overdue reservations.

While e-RoomReserve does not directly control building systems such as HVAC or lighting, the system's BLE-based occupancy detection and automated staff notification mechanisms provide the real-time visibility necessary for informed facility management decisions. By alerting staff when rooms become vacant ahead of schedule or when occupancy persists beyond reserved periods, e-RoomReserve supports more responsive and efficient room management — reducing unnecessary facility usage and improving the accuracy of occupancy records available to administrators for planning and reporting purposes.

2.6 System Architecture and Security in IoT Monitoring Systems

The design of IoT-based monitoring systems requires careful consideration of both architectural structure and security mechanisms, as the effectiveness of the system depends not only on the accuracy of its sensing components but also on the reliability of data transmission, the integrity of stored information, and the protection of user privacy. A well-designed IoT architecture typically follows a layered model — from physical sensing devices, through communication networks, to cloud-based data processing and user-facing application interfaces.

Chaudhari et al. (2024) describe the standard IoT architecture for smart building systems as consisting of sensors collecting environmental and occupancy data, a data fusion center that processes and analyzes incoming readings, and a control system that issues commands to building equipment based on the processed data. This layered approach — where each component performs a distinct function while remaining interconnected — directly informs the architecture of e-RoomReserve, which follows a similar structure: ESP32 BLE beacon modules serve as the sensing layer, the mobile application and Next.js server handle communication and processing, Firebase provides cloud-based data storage and real-time synchronization, and the administrative portal serves as the user-facing control interface.

The specific use of Firebase as a real-time cloud backend for IoT monitoring systems is validated by Le et al. (2021), who developed a real-time IoT monitoring system using a Firebase database to store and share environmental sensor data, paired with a mobile application that allows users to monitor sensor readings and control actuators remotely over the internet. Their

system architecture — comprising sensor nodes, an IoT gateway, a Firebase real-time database, and a mobile application — closely parallels the component structure of e-RoomReserve. Le et al. (2021) also incorporated a built-in hardware security function using AES128 encryption to protect data transmitted between sensor nodes and the cloud backend, highlighting that security considerations must be addressed at both the hardware and software levels of IoT system design.

Security and privacy concerns are particularly significant in occupancy monitoring systems, where the collection of presence data and device identifiers raises questions about user consent and data protection. Franco et al. (2025) address this directly in the context of a university IoT deployment, emphasizing that IoT networks are intrinsically less secure than wired networks and that cybersecurity must be treated as a priority. Their study adopted a Zero Trust Architecture approach — which enforces security based on the trustworthiness of connecting entities rather than relying on secure perimeters — and restricted network access to authenticated sensors and authorized users only. These principles are reflected in e-RoomReserve's design, which incorporates role-based authentication and authorization, data anonymization of BLE device identifiers, and explicit user consent mechanisms as described in the system's scope and limitations.

At a broader level, Olimat et al. (2023) note that despite the growing adoption of IoT technologies in facility management, interoperability between different systems and the secure exchange of data between platforms remain significant challenges. Their analysis underscores the importance of designing IoT systems with standardized communication protocols and clear data management policies — considerations that e-RoomReserve addresses through its use of Firebase's structured real-time database, RESTful API endpoints via Next.js, and clearly defined user roles for students, faculty, and administrators.

2.7 Smart Campus Monitoring in the Philippine Context

The adoption of IoT technologies in Philippine higher education institutions has gained increasing attention in recent years, driven by the recognition that digital transformation in campus operations can significantly improve facility management, resource utilization, and administrative efficiency. While smart campus initiatives have been widely studied in international contexts, the Philippine educational landscape presents distinct challenges —

including limited infrastructure, varying levels of digital readiness across institutions, and a historically strong reliance on manual processes for facility management and room scheduling.

Taruc and De La Cruz (2024) conducted a research analysis of NB-IoT and IoT use cases in universities, campuses, and educational institutions, identifying six key application areas including smart campus management, asset tracking, environmental monitoring, and safety and security systems. Their study concludes that IoT technology has significant potential to enhance various aspects of educational institutions in the Philippines, from improving operational efficiency to strengthening campus security — establishing a broad empirical foundation for IoT adoption in the local higher education context. The study's findings situate e-RoomReserve within a growing body of Philippine IoT research and validate the relevance of applying BLE-based monitoring and digital reservation technology to local campus environments.

At the institutional level, the Department of Science and Technology (2025) has actively promoted IoT integration in Philippine educational institutions through training initiatives in robotics, AI, and IoT, signaling national-level recognition of the importance of smart technology adoption in schools and universities. This government-backed push for digital transformation in education underscores the timeliness and alignment of e-RoomReserve with broader national priorities for modernizing educational infrastructure.

The specific challenge of room reservation management in Philippine higher education institutions is directly addressed by German et al. (2021), who analyzed and optimized the facility reservation process of a Philippine higher education institution where manual coordination across multiple campus entities required significant traveling activities and created persistent issues with the availability of authorized personnel responsible for granting facility use. Through information system design, their study developed an online resource and room management system to handle real-time information exchange and minimize the effort and time required to complete a reservation. The problems documented by German et al. (2021) — fragmented manual processes, lack of real-time availability information, and administrative inefficiency — closely mirror the conditions described in Chapter 1 of this study at St. Dominic College of Asia, reinforcing the need for an integrated digital solution such as e-RoomReserve that addresses these persistent institutional challenges through automated monitoring, structured approval workflows, and real-time occupancy detection.

Collectively, these studies establish that the development of e-RoomReserve is both timely and contextually grounded, responding to documented gaps in smart campus adoption and room management efficiency within the Philippine higher education sector.

2.8 Research Gap

Recent studies confirm that IoT monitoring, BLE-based presence detection, digital reservation platforms, and AI-based recommendation systems are each effective individually in improving facility management and space utilization. Chaudhari et al. (2024) and Khan et al. (2024) demonstrate the effectiveness of IoT-based occupancy detection in smart buildings, while Demrozi et al. (2021) and Oh (2021) validate BLE beacons as a practical and low-cost approach to indoor presence monitoring. Zakarija et al. (2025) and Somwong et al. (2022) confirm that digital reservation platforms improve scheduling efficiency, and Ricci et al. (2022) and Syazwani et al. (2020) establish the theoretical and practical foundations for AI-based recommendation and feedback analysis in institutional systems.

However, most existing implementations focus on isolated system components rather than integrating all of these functions into a single unified platform. Few studies combine real-time BLE-based occupancy detection with structured reservation workflows, faculty approval mechanisms, AI-based room recommendations, and post-use feedback analysis within one cohesive system. Furthermore, as documented by German et al. (2021) and Taruc and De La Cruz (2024), many Philippine higher education institutions continue to rely on manual reservation and monitoring processes, leaving a significant gap between the capabilities demonstrated in international research and the operational reality of local campus environments.

The present study addresses these gaps by developing e-RoomReserve — an AI-integrated room reservation and occupancy monitoring system that combines BLE beacon-based presence detection, a structured faculty approval workflow, real-time occupancy monitoring, an AI-based room recommendation module with content-based filtering, and a sentiment analysis-based feedback system — within a single platform designed specifically for the institutional context of St. Dominic College of Asia.

2.9 Synthesis of Related Literature

The reviewed literature establishes a strong theoretical and empirical foundation for the design and development of e-RoomReserve across all of its core components. Studies on IoT in smart buildings consistently demonstrate that real-time occupancy data improves operational efficiency and facility management — with Franco et al. (2025) and Chaudhari et al. (2024) specifically documenting improvements in space utilization and operational decision-making in university buildings through real-time occupancy monitoring. These findings directly support e-RoomReserve's occupancy monitoring and staff notification mechanisms.

Research on BLE beacon technology confirms its suitability for room-level presence detection in indoor environments. Demrozi et al. (2021) validate low-cost BLE devices for occupancy estimation in public buildings, Oh (2021) demonstrates the applicability of BLE beacon modules in real-world indoor spaces with obstacles, and Janczak et al. (2022) provide a detailed analysis of RSSI accuracy and its limitations under varying environmental conditions — grounding e-RoomReserve's known detection constraints in established literature.

Studies on digital reservation systems highlight that effective platforms must go beyond simple scheduling to incorporate real-time occupancy verification and structured administrative workflows. Zakarija et al. (2025) and Somwong et al. (2022) demonstrate that integrating occupancy detection with booking interfaces significantly improves space utilization and reduces the gap between scheduled and actual room usage — a core operational goal of e-RoomReserve.

The AI-based recommendation and feedback components of e-RoomReserve are grounded in the foundational work of Ricci et al. (2022) on content-based filtering and recommender system design, and validated by Syazwani et al. (2020), whose study demonstrates that VADER sentiment analysis achieves approximately 90% accuracy when applied to informal student feedback in educational contexts — supporting its selection for processing post-reservation survey responses.

At the system architecture level, Le et al. (2021) validate Firebase as a reliable real-time backend for IoT monitoring systems with built-in security mechanisms, while Franco et al. (2025) and Olimat et al. (2023) address the broader cybersecurity and interoperability challenges inherent in IoT-based institutional deployments.

Finally, in the Philippine context, Taruc and De La Cruz (2024) and German et al. (2021) confirm that IoT adoption in local higher education institutions is both feasible and urgently needed, with existing manual processes creating persistent inefficiencies in room scheduling and facility management that an integrated digital system such as e-RoomReserve is designed to address.

Taken together, the reviewed literature demonstrates that while individual technologies — BLE detection, digital reservation platforms, AI-based recommendation, and IoT architecture — have each been validated in isolation, no existing study integrates all of these components into a single system tailored to the Philippine higher education context. e-RoomReserve represents a direct and contextually grounded response to this gap.

Chapter III

Project Design and Development

3.1 System Objective

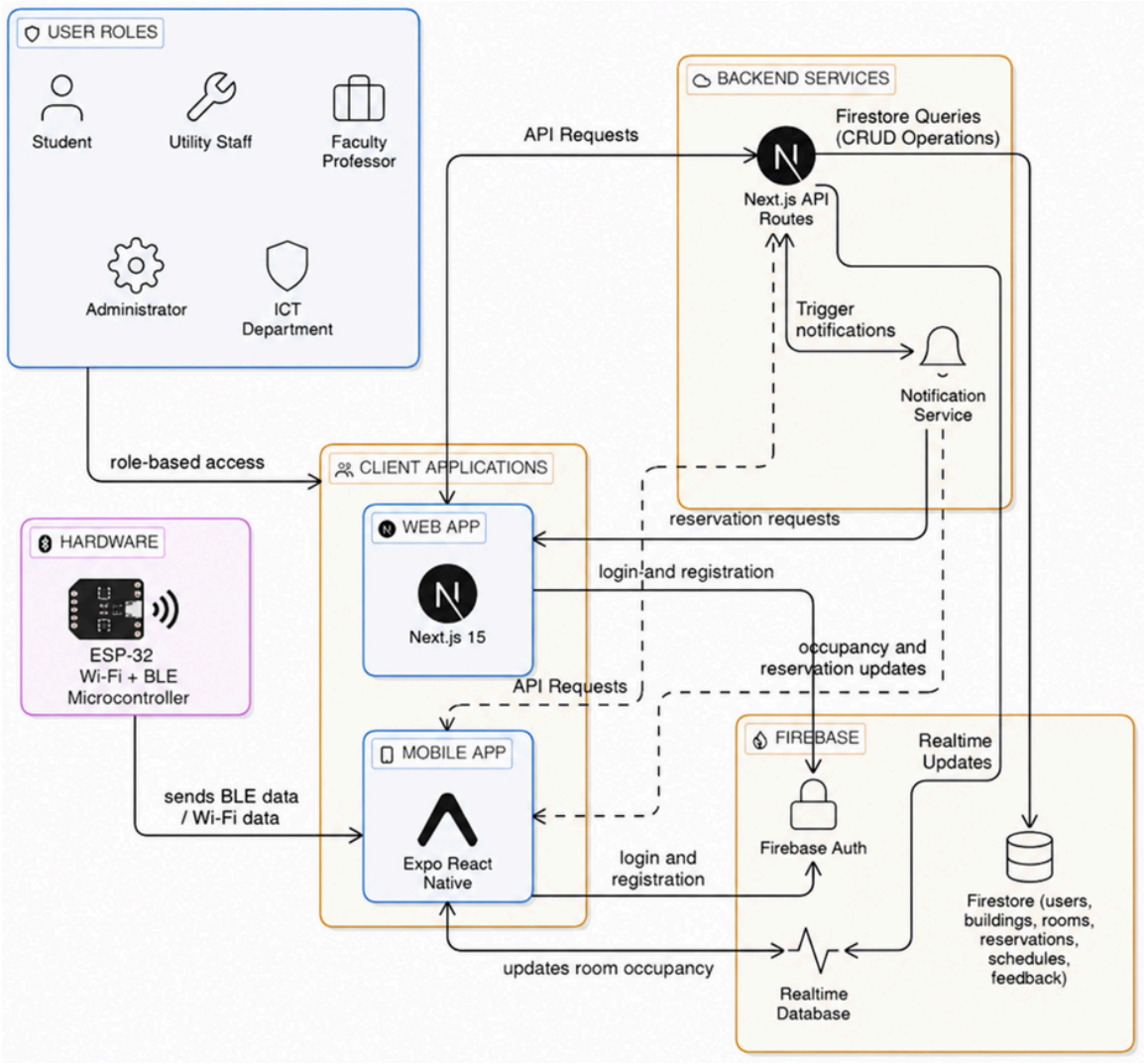
The primary objective of this project is to design and develop an AI-Integrated Room Reservation and Occupancy Monitoring System using Bluetooth Low Energy (BLE) beacon technology for St. Dominic College of Asia. The system aims to address existing challenges in manual room scheduling, lack of occupancy verification, and inefficient facility utilization by providing a centralized and automated digital platform.

To support the formal objectives of the study, the system is designed to achieve the following operational goals:

1. Provide a centralized platform for submitting paperless room reservation requests, processing faculty approvals, and monitoring real-time room availability.
2. Implement BLE-based automatic occupancy detection through device proximity scanning using the ESP32 BLE Beacon Module.
3. Utilize BLE beacon technology to continuously monitor room occupancy without requiring manual user interaction.
4. Display real-time room status through a web-based dashboard and mobile application accessible to users and administrators.
5. Reduce scheduling conflicts, unauthorized room usage, and administrative workload through automation.

The system is designed to improve operational efficiency, transparency, and resource management while supporting future smart campus initiatives at St. Dominic College of Asia.

3.2 System Framework



The system framework of the e-RoomReserve is a multi-layered architecture that integrates client applications, backend services, cloud infrastructure, and hardware components to enable intelligent room reservation and occupancy monitoring. Users—including students, faculty professors, utility staff, administrators, and the ICT department—interact with the system through either a web application built with Next.js or a mobile application developed using Expo React Native. Access to features is governed through role-based permissions, ensuring that each user type can only perform actions according to their responsibilities, such as making reservations, approving requests, or managing facilities.

Authentication is handled directly by Firebase Authentication, where both web and mobile clients perform login and registration using email/password or Google-based authentication. Once authenticated, users interact with the system's core functionalities through API requests sent to Next.js API routes, which act as the backend service layer. These API routes handle business logic such as reservation processing, room management, and scheduling, and communicate with Firestore to perform structured data operations (CRUD) on collections like users, rooms, buildings, and reservations. For real-time features, such as room occupancy updates, the system utilizes Firebase Realtime Database, enabling instant synchronization of room status across clients.

A key feature of the system is its integration with ESP32 BLE Beacon Module installed in rooms. These beacons continuously broadcast unique identifiers, which are detected by the mobile application. When a user with an active reservation is within range, the mobile app interprets the BLE signal and updates the room's occupancy status in the Realtime Database. This allows the system to reflect real-time usage of rooms and automate processes such as notifying utility staff when a reservation begins or ends.

The backend also includes a notification service that is triggered by the API layer based on reservation events and occupancy changes. This service sends alerts to users—particularly utility staff and faculty—ensuring timely awareness of room usage and scheduling changes. Overall, the framework demonstrates a clear separation of concerns: client applications handle user interaction and authentication, the API layer manages business logic, Firebase provides scalable cloud data services, and BLE hardware enables real-world context awareness, resulting in a cohesive and responsive smart reservation system.

1. Presentation Layer

The Presentation Layer serves as the primary interface through which users interact with the e-RoomReserve system. It is implemented through two client-facing applications: a mobile application built with React Native and Expo for students and faculty to submit reservation requests and receive real-time notifications, and a web-based administrative portal developed using Next.js for authorized personnel to manage approvals, monitor occupancy, and view

analytics. Both client applications communicate with the Application Layer through RESTful API endpoints, ensuring a consistent and responsive user experience across platforms.

This layer supports the following user-facing functions:

- Submit room reservation requests via the mobile application (students and faculty)
- View real-time room availability, occupancy status, and reservation history
- Track reservation progress, approval status, and push notification alerts
- Access the web-based administrative dashboard for reservation approvals, occupancy monitoring, and usage reporting (authorized personnel only)

It captures all user inputs from both the mobile and web interfaces, and renders real-time system outputs including occupancy status, approval decisions, and automated notifications.

2. Application Layer

The Application Layer contains the core business logic and processing mechanisms of e-RoomReserve. It is implemented as a server-side layer using Next.js API routes, which expose RESTful endpoints consumed by both the mobile application and the administrative portal. This layer mediates all interactions between the Presentation Layer and the Data Layer, ensuring that every request is validated, authorized, and correctly processed before data is read or written.

It is responsible for the following core functions:

- User authentication and role-based access control, distinguishing between Student, Faculty, and Administrator roles
- Reservation workflow management, including request submission, faculty approval or rejection, conflict detection, and automated vacancy release
- BLE-based occupancy detection logic, including RSSI signal evaluation, configurable timeout processing, and human-in-the-loop fallback triggering when signal loss is detected
- Automated push notifications and staff alerts via Firebase Cloud Messaging, triggered by reservation events, approval decisions, and occupancy changes

- Data processing and validation, including AI-based room recommendation using content-based filtering and sentiment analysis of post-reservation survey responses via VADER

This layer acts as the system's processing engine, coordinating all inputs from both user interactions and BLE device signals, and orchestrating the appropriate responses across all connected layers.

3. Data Layer

The Data Layer is powered by a combination of Firebase and Supabase, which together provide a comprehensive suite of cloud-based services responsible for persistent data storage, file uploads, real-time synchronization, authentication management, and push notification delivery. All data generated by both the Presentation Layer and the Application Layer is ultimately stored, retrieved, and synchronized through this layer.

The Data Layer comprises the following Firebase services:

- Firebase Firestore for structured, persistent storage of user profiles, room records, reservation requests, occupancy logs, and post-reservation survey responses
- Firebase Authentication for secure user login, session management, and role-based identity verification across student, faculty, and administrator accounts
- Firebase Cloud Messaging (FCM) for delivering push notifications to mobile devices and web browsers, supporting reservation status updates and staff alert messages
- Firebase Realtime Database for low-latency live data synchronization, enabling instantaneous occupancy status updates to be reflected across all connected client applications
- Supabase Storage for managing file and image uploads, including room profile photos and reservation-related attachments, complementing Firebase services with dedicated object storage capabilities

This layer ensures reliable, real-time data availability across all connected client applications and system components, while providing the persistence and integrity required for reporting, audit trails, and facility management analytics.

4. Device Layer

The Device Layer represents the physical hardware tier of the e-RoomReserve architecture. It consists of the ESP32 BLE Beacon Module installed in each monitored room, as well as the smartphones of registered users running the React Native mobile application. This layer is the source of all real-time occupancy data in the system and operates independently of user-initiated actions, enabling passive and continuous room monitoring.

- ESP32 BLE Beacon Module are physically mounted inside each designated room, configured to broadcast iBeacon advertisement packets at regular intervals
- Each beacon continuously broadcasts a unique UUID along with Major and Minor values, which are detected by the mobile application running on the user's smartphone when within proximity range
- The ESP32 beacon detects BLE client connection and disconnection events, posting occupancy payloads to the Next.js backend via HTTP. The mobile application uses RSSI only during the reservation activation phase to verify that the user is physically near the room before initiating a reservation. Once a reservation is active, occupancy is determined by the ESP32's BLE connection state rather than continuous RSSI estimation, with staff physically verifying room status before any vacancy changes are confirmed

This layer enables fully automated, real-time occupancy monitoring without requiring any manual check-in action from the user. It operates continuously in the background as long as the mobile application is active and Bluetooth is enabled on the user's device, providing the foundational sensing capability upon which the entire occupancy detection system depends.

System Data Flow

The system follows a four-stage data flow that connects all layers in a continuous cycle:

1. User inputs (reservation requests, approval decisions, and feedback responses) are captured through the Presentation Layer via the mobile application or the administrative web portal

2. The Application Layer processes these inputs alongside BLE signal data from user devices, applying business logic for reservation management, conflict detection, occupancy evaluation, and notification dispatch
3. The Device Layer continuously provides real-time occupancy data through the ESP32 BLE Beacon Module, which detects BLE client connection and disconnection events and posts occupancy payloads directly to the Next.js backend via HTTP
4. The Data Layer stores all processed data in Firebase, synchronizing reservation states, occupancy records, and notification events in real time back to the Presentation Layer for display to users and administrators

This end-to-end flow ensures accurate, real-time room monitoring and responsive system behavior across all user roles and interaction points.

3.3 Feasibility Study

A feasibility study was conducted to evaluate whether e-RoomReserve can be successfully designed, developed, and deployed within the operational, technical, and financial constraints of St. Dominic College of Asia. The study examines the project across three dimensions: technical feasibility, which assesses the availability and suitability of the required technologies; operational feasibility, which determines whether the system can be effectively integrated into existing institutional workflows; and economic feasibility, which evaluates the projected costs and potential return on investment. The findings from each dimension confirm that the project is viable and appropriate for institutional implementation.

3.3.1 Technical Feasibility

The project is technically feasible due to the availability, compatibility, and proven effectiveness of the modern web, mobile, and IoT technologies selected for e-RoomReserve.

- **Next.js** provides a full-stack development framework supporting both server-side rendering for the administrative web portal and RESTful API routes for backend business logic, eliminating the need for a separate backend server

- **React Native with Expo** enables cross-platform mobile application development for both iOS and Android devices using a single JavaScript codebase, with built-in support for Bluetooth Low Energy scanning through available Expo and React Native BLE libraries
- **Firebase** offers a comprehensive suite of cloud services including real-time database synchronization, structured data storage via Firestore, user authentication, and push notification delivery through Firebase Cloud Messaging — all available under a free-tier plan suitable for prototype deployment
- **Supabase** provides open-source object storage for file and image uploads, including room profile photos and reservation-related attachments — available under a free-tier plan compatible with prototype deployment and integrating seamlessly alongside Firebase services
- **ESP32 BLE Beacon Modules** are low-cost, widely available hardware components that are flashed with e-RoomReserve firmware via the Arduino IDE, with room-specific values configured including roomId, roomName, beaconId, Wi-Fi credentials, and the Next.js server URL before installation. Each beacon unit is housed in a protective enclosure box for secure room installation and powered through a USB adapter for continuous, reliable operation without battery dependency
- **VADER (Valence Aware Dictionary and sEntiment Reasoner)** is a free, open-source sentiment analysis library available in the JavaScript ecosystem via npm, enabling lightweight sentiment classification of post-reservation survey responses without requiring external AI services, model training, or additional infrastructure

All selected technologies are well-documented, open-source or freely available, and compatible with existing smartphone and cloud infrastructure, making the system technically viable for prototype implementation and future campus-wide deployment.

3.3.2 Operational Feasibility

The system is operationally feasible as it enhances existing processes rather than replacing them entirely.

- The reservation system follows familiar online booking workflows
- BLE-based detection eliminates the need for manual check-in
- The administrative dashboard simplifies approval and monitoring tasks

Minimal training is required for users, making the system easy to adopt for students, faculty, and administrators.

3.3.3 Economic Feasibility

The project requires a low initial cost due to the use of affordable IoT hardware and free software platforms.

Estimated Cost per Room

Component	Description	Estimated Cost (PHP)
ESP32 BLE Beacon Module	Room-mounted beacon for occupancy detection	₱80 – ₱150
USB Power Adapter	Continuous USB power source for beacon operation	₱50 – ₱150
Protective Enclosure Box	Protective plastic enclosure box for secure beacon mounting inside rooms	₱50 – ₱100
Miscellaneous	Cables and connectors	₱50 – ₱100
Total		**₱230 - ₱500**

Software costs are minimal:

- Next.js is open-source and free
- React Native and Expo are open-source and free
- Firebase provides a free-tier plan suitable for prototype deployment

- Supabase provides a free-tier plan that includes object storage suitable for prototype deployment alongside Firebase
- VADER is a free, open-source library available via npm

Payback Period

The primary return on investment for e-RoomReserve comes from operational efficiency gains. By reducing administrative workload, minimizing scheduling conflicts, and eliminating manual room monitoring rounds, the system reduces the time and labor costs associated with the current manual reservation and inspection processes at St. Dominic College of Asia. Hardware costs per room are estimated at ₱230 to ₱500, which is recoverable through sustained improvements in facility utilization and staff productivity over time.

3.4 System Design – System Development Life Cycle (SDLC)

The project follows the **Agile Methodology using the Scrum Framework**.

This approach is chosen because:

- The system involves both hardware and software integration
- Iterative development allows continuous testing and improvement
- It enables flexibility in handling changes during development

Sprint Plan

Sprint	Duration	Key Tasks	Deliverable
Sprint 0	Week 1–2	Setup, planning, wireframes	Project setup
Sprint 1	Week 3–4	Authentication system	Login module
Sprint 2	Week 5–6	Reservation system	Booking module
Sprint 3	Week 7–8	BLE detection, occupancy logic	Occupancy system

Sprint 4	Week 9–10	Alerts, dashboard, AI recommendation module, sentiment analysis integration	Monitoring system
Sprint 5	Week 11–12	Testing, integration	Final system
Sprint 6	Week 13	BLE beacon installation, cloud deployment, user training, UAT	Deployed system

Scrum Activities

- Sprint Planning
- Daily Stand-up
- Sprint Review
- Sprint Retrospective

Each sprint produces a working and testable system component.

3.4.1 User Interface

The system provides two distinct user-facing interfaces tailored to different user roles and interaction contexts.

The **mobile application**, developed using React Native and Expo, is the primary interface for students and faculty making reservations. It provides access to the following features:

- User login and registration
- Room reservation request submission
- Real-time room availability browsing
- AI-based alternative room recommendation display when preferred room is unavailable
- BLE beacon scanning status and occupancy detection indicator
- Bluetooth enable prompt upon app launch and before check-in

- Reservation status tracking and approval notifications
- Post-reservation survey submission

The **administrative web portal**, developed using Next.js, is accessible to authorized faculty, utility staff, and administrators through any modern desktop or mobile browser. It provides access to the following features:

- Faculty reservation approval and rejection panel
- Real-time occupancy dashboard
- Staff alert notifications for BLE signal loss and overtime occupancy
- Room condition and user satisfaction reports generated from post-reservation surveys
- Sentiment analysis insights and auto-generated facility maintenance flags
- Usage analytics and room utilization reports
- Room and building management

Both interfaces are designed to be simple, intuitive, and user-friendly, with role-based access control ensuring that each user type only sees features relevant to their responsibilities.

3.4.2 System Input and Output

System Inputs

- User login credentials for authentication
- Room reservation request details including date, time, room selection, and purpose
- Faculty approval or rejection decisions on pending reservation requests
- BLE signal strength (RSSI) readings from ESP32 BLE Beacon Module detected by the mobile application
- Bluetooth enable/disable status from the user's mobile device
- Post-reservation survey responses from students including satisfaction ratings and free-text comments
- Post-reservation staff survey responses including room condition flags and free-text notes

System Outputs

- Reservation confirmation, approval, or rejection notifications via push notification
- Real-time room occupancy status updates across all connected client applications
- Automated staff alerts for BLE signal loss during active reservations, overtime occupancy, and required room inspection
- AI-generated alternative room recommendations based on room attributes and availability
- Sentiment analysis classification of free-text survey responses as positive, negative, or neutral
- Auto-generated facility insights for faculty and administrators such as recurring cleanliness issues or equipment condition flags
- Occupancy logs and room utilization analytics reports
- Vacancy status notifications triggering staff to deactivate room facilities

3.5 Gantt Chart

The project spans approximately thirteen weeks, structured into seven Agile sprints following the Scrum framework. The Gantt chart below maps each sprint and its key tasks across the project timeline, from initial setup through to final deployment. Each sprint follows a two-week cycle except Sprint 6, which is a dedicated one-week deployment sprint. The timeline accommodates iterative development, continuous testing, and user acceptance evaluation prior to full deployment.

Each sprint includes:

- Planning
- Development
- Review
- Retrospective

The Gantt chart below illustrates the project schedule across all 13 weeks. Shaded cells indicate the weeks during which each sprint's tasks are actively being executed.

Sprint / Task	W 1	W 2	W 3	W 4	W 5	W 6	W 7	W 8	W 9	W 10	W 11	W 12	W 13
Sprint 0 – Setup & Planning													
Sprint 1 – Authentication													
Sprint 2 – Reservation System													
Sprint 3 – BLE Detection, Occupancy Logic													
Sprint 4 – Alerts, Dashboard, AI Recommendation Module, Sentiment Analysis													
Sprint 5 – Testing & Integration													
Sprint 6 – Deployment													

3.6 Testing Process

The system undergoes three levels of testing to validate functionality, integration, and user acceptance prior to deployment.

3.6.1 Unit and System Testing

Each module is tested independently to verify individual functionality, followed by end-to-end system testing that evaluates the complete workflow from reservation submission through BLE occupancy detection and staff notification. This includes testing BLE signal detection accuracy, timeout logic, Firebase synchronization, push notifications, AI room recommendation output, and sentiment analysis classification.

3.6.2 Integration Testing

All system components — the mobile application, administrative web portal, Next.js backend, Firebase services, and ESP32 BLE Beacon Module — are tested together to verify that data flows correctly across all layers and that real-time occupancy updates, reservation workflows, and alert mechanisms function as expected in an integrated environment.

3.6.3 User Acceptance Testing (UAT)

Selected students, faculty, and staff test the system in a real classroom environment based on the following criteria:

- Usability — ease of use across both the mobile application and administrative portal
- Reliability — consistency of BLE occupancy detection and reservation workflows
- Performance — responsiveness of the system within acceptable thresholds
- Security — correctness of role-based access control and data privacy measures

3.6.4 Performance Benchmarks

Indicator	Acceptable Threshold
Occupancy detection accuracy	≥ 90%
API response time	≤ 2 seconds
Dashboard update latency	≤ 3 seconds
Conflict detection	100%
BLE detection response time	≤ 5 seconds
Sentiment analysis accuracy	≥ 85%

3.7 Installation and Deployment

Deployment follows a phased approach to ensure stable implementation and minimize disruption to campus operations.

Phase 1: Pilot Deployment

- ESP32 BLE Beacon Module units flashed with e-RoomReserve firmware via the Arduino IDE, with room-specific values configured including roomId, roomName, beaconId, Wi-Fi credentials, and the Next.js server URL before installation
- Beacons physically mounted inside selected classrooms at optimal positions for signal coverage
- Mobile application deployed through Expo for iOS and Android devices
- Administrative web portal deployed via cloud hosting on Vercel
- Firebase backend configured with production security rules and role-based access control
- Users trained on mobile application usage, Bluetooth requirements, and reservation workflows

Phase 2: Expansion

- Additional classrooms integrated with ESP32 BLE Beacon Module
- System calibrated based on pilot feedback and RSSI threshold adjustments
- Performance improvements and feature refinements applied based on UAT results
- Campus-wide deployment planned following successful pilot evaluation

3.8 Maintenance

The system requires ongoing maintenance to ensure long-term performance, reliability, and security.

Maintenance Activities

- System monitoring and error tracking through Firebase and Vercel dashboards
- Software updates and security patches for Next.js, React Native Expo, and Firebase dependencies
- Database optimization including Firestore index management and Realtime Database cleanup of outdated occupancy records
- Supabase Storage monitoring including periodic review of uploaded file sizes, bucket usage, and access logs to ensure storage limits remain within the free-tier plan during prototype deployment

- ESP32 BLE Beacon Module hardware inspection and power supply monitoring — USB adapters should be inspected periodically to ensure continuous power delivery to beacon units; enclosure boxes should be checked for physical damage or displacement during scheduled maintenance rounds
- RSSI threshold recalibration if room layouts or furniture arrangements change significantly
- Sentiment analysis model monitoring to ensure VADER classification accuracy remains acceptable as feedback volume grows
- System scalability assessment as the number of monitored rooms increases

User feedback collected through post-reservation surveys is continuously analyzed to identify recurring issues and inform future system improvements.

CHAPTER IV
PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA

This chapter presents the developed e-RoomReserve system, the results of the testing conducted in accordance with Section 3.6, and the findings of the expert and end-user evaluations. The evaluation results are interpreted using a five-point Likert scale, organized according to the eight software product quality characteristics of the ISO/IEC 25010 model (functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability), and discussed in relation to the objectives of the study and the related literature reviewed in Chapter II.

4.1 Presentation of the Developed System

e-RoomReserve was developed as an integrated system of three parts: a Next.js web application that also serves as the backend API, an Expo React Native mobile application, and ESP32 BLE beacon firmware. Firebase Authentication and Firestore serve as the core identity and data services, Supabase Storage holds uploaded reservation documents, and push notifications are delivered through Expo. The final implementation follows the four-layer framework described in Section 3.2 and is summarized below.

4.1.1 Web Application

The web application provides registration and login through email/password (with email verification) or Google Sign-In, followed by role-based dashboards. Students and faculty professors can reserve rooms, track reservations, submit feedback, use the inbox, view room status, and perform Bluetooth check-in through Web Bluetooth on supported browsers. Administrators manage rooms, buildings, class schedules, pending reservations,

and feedback, and monitor BLE beacon status for their assigned buildings. Super administrators approve and manage higher-level accounts. The administrator Feedback tab presents sentiment analytics by reporting period (all time, weekly, monthly, semester, or yearly), ranked positive and negative aspects, trend insights, and an optional gender comparison.

[Insert screenshot: Web dashboard (Administrator view showing room status and pending approvals)]

Figure 4.1 e-RoomReserve Web Application Administrator Dashboard

4.1.2 Mobile Application

The mobile application allows users to select a campus (Main or Digital), browse buildings, floors, and rooms, and submit reservations through the reservation form. When the preferred room is unavailable, an alternative-rooms screen suggests other available rooms. Users can view their reservation history and inbox, and can perform Bluetooth check-in by scanning for the room's ESP32 beacon and matching its beacon ID against the reserved room. After check-in, an ongoing presence monitor sends a heartbeat every 30 seconds, performs a 10-second scan pass, and uses a default RSSI threshold of -75; two consecutive beacon warnings are required before a transient problem is surfaced. Utility staff are taken directly to a dashboard listing active and awaiting-release reservations for their campus, where they confirm physical vacancy before a room is released.

[Insert screenshot: Mobile reservation form, Bluetooth check-in, and reservation tracking screens]

Figure 4.2 e-RoomReserve Mobile Reservation and Bluetooth Check-In Interface

4.1.3 BLE Beacon Hardware

Each participating room hosts an ESP32 unit running NimBLE firmware. The unit uses the beacon ID as its BLE device name, advertises a service UUID, and exposes a beacon ID characteristic and a room ID characteristic. Through Wi-Fi, it posts telemetry (room ID, room name, beacon ID, occupancy, timestamp, connection status, and event type) to the occupancy endpoint every 30 seconds and whenever a device connects or disconnects. The event types observed are DEVICE_CONNECTED, DEVICE_DISCONNECTED, INTERVAL_UPDATE, and END_OF_RESERVATION. The backend stores the latest beacon connection metadata on the corresponding room document in Firestore.

[Insert screenshot: Installed ESP32 beacon unit in its protective enclosure]

Figure 4.3 Installed ESP32 BLE Beacon Unit

4.1.4 Reservation, Approval, and Occupancy Workflow

The reservation lifecycle implemented in the system proceeds as follows:

1. The user submits a reservation. The backend validates the request against class schedules and existing reservations, and rejects conflicting time slots.
2. An approval flow is generated according to role and campus. Faculty reservations skip the adviser step and go to the building administrator; Digital Campus reservations use the building administrator flow; Main Campus student reservations require an adviser (or department head/professor) and the building administrator. Approver accounts and uploaded approval documents are validated.

3. Once approved, the user checks in by Bluetooth while physically near the room beacon. The backend records the check-in method and time and marks the room as occupied.
4. The presence monitor sends periodic heartbeats. The server marks presence as healthy, warning, timed out, or stopped, and staff are alerted when signal loss is detected.
5. When the session ends, the reservation owner marks it as completed, which stops presence monitoring immediately. Utility staff then physically confirm that the room is vacant before the occupancy release is recorded and the room history is updated.

4.1.5 Feedback Analytics, Bulk Schedule Import, and DC Space Integration

After a reservation, users submit a 1-5 overall rating, five category ratings (cleanliness, comfort, air conditioning, equipment/projector, and internet connectivity), and free-text comments. The backend applies VADER to compute positive, neutral, negative, and compound scores, classifies the result into five sentiment bands, detects room-related aspects, and extracts keywords. Sentiment metrics for any gender group with fewer than five samples are withheld to protect small groups.

The bulk schedule import allows building administrators to upload Excel workbooks (.xlsx or .xls) in either list or timetable layout. The file is parsed in the browser and shown as a preview in which each detected class block is matched to a room and validated for required fields, time ranges, and overlaps with other rows or existing schedules. Only valid rows are imported by default, and conflicts with existing schedules can be overridden after confirmation. Each imported row is saved through the standard schedule API, so server-side authorization and conflict checks still apply.

For event-related reservations, DC Space and e-RoomReserve exchange data through an authenticated server-to-server integration. DC Space submits the event details to a protected endpoint using a Bearer secret, and the DC Space event ID serves as an idempotency key so that repeated deliveries do not create duplicate reservations. The linked reservation goes through the normal conflict and approval rules and is marked as an event. Status changes (approved, rejected, cancelled, and completed) are sent back to DC Space through a configured callback, and a failed delivery does not undo the reservation update. The two systems still keep separate databases.

[Insert screenshot: Bulk schedule import preview with validation results, and Feedback analytics dashboard]

Figure 4.4 Bulk Schedule Import Preview and Feedback Sentiment Analytics

4.1.6 Differences from the Proposed Design

Two features were implemented differently from the description in Chapters I to III. First, the bulk schedule import accepts Excel workbooks (.xlsx and .xls) that are parsed in the browser, rather than a generic CSV or Excel file uploaded to the server. Second, the link to DC Space was implemented as an authenticated server-to-server integration with status callbacks, going beyond the passive reference link originally described, while the two systems continue to maintain separate databases. A button in My Reservations that opens the deployed DC Space application for event reservations is planned but not yet implemented.

4.2 Profile of the Respondents

The main end-users of e-RoomReserve are the officers of student organizations, who most frequently reserve campus facilities. St. Dominic College

of Asia has 23 recognized organizations, each with at least seven officers, which gives a population of 161 organization officers (23 x 7). Using a sample size calculator at a 95% confidence level and a 5% margin of error, the required sample size is 114 respondents. These 114 respondents consist of 92 end-users, who are organization officers, and 22 IT experts. The end-users were selected through [sampling technique, e.g., stratified random sampling per organization], and the experts through [purposive sampling]. The evaluation took place from [start date] to [end date] following the user acceptance testing described in Section 3.6.3. Tables 4.1 and 4.2 present the profile of each group.

Profile Variable	Category	Frequency	Percentage
Position/Designation	[e.g., IT Faculty]	[f]	[%]
	[e.g., Software Developer]	[f]	[%]
	[e.g., Systems/Network Administrator]	[f]	[%]
Field of Specialization	[e.g., Software/Web Development]	[f]	[%]
	[e.g., Networking/IoT]	[f]	[%]
	[e.g., Database/Systems Analysis]	[f]	[%]
Years of Experience in IT	[1-5 years]	[f]	[%]
	[6-10 years]	[f]	[%]
	[11 years or more]	[f]	[%]
Total		**22**	**100%**

Table 4.1 Profile of Expert Respondents

Profile Variable	Category	Frequency	Percentage
Role	Student officer	[f]	[%]
	Faculty (organization adviser)	[f]	[%]

Profile Variable	Category	Frequency	Percentage
College/Department of Organization	[Organization / College 1]	[f]	[%]
	[Organization / College 2]	[f]	[%]
	[Organization / College 3]	[f]	[%]
Expected Frequency of Use	At least once a week	[f]	[%]
	A few times a month	[f]	[%]
	Once a month or less	[f]	[%]
Total		**92**	**100%**

Table 4.2 Profile of End-User Respondents

4.3 Results of System Testing

Testing followed the three levels defined in Section 3.6: unit and system testing, integration testing, and performance benchmarking, with user acceptance testing reported through the evaluation results in Sections 4.5 and 4.6. Testing was carried out in [number] pilot rooms located in [building/s].

4.3.1 Unit and System Testing

Each module was tested independently, followed by end-to-end testing from reservation submission to BLE check-in, presence monitoring, and staff release. Of the 16 test cases executed, [n] ([%]) passed on the final run. Table 4.3 summarizes the test cases and their outcomes. Defects found in earlier runs, such as [describe notable defect and fix], were corrected during Sprint 5.

St. Dominic Complex, E. Aguinaldo Highway, Bacoor City, Cavite, Philippines 4102
0917 625 8222 • 0922 850 9200 • (046) 417 7322 • (046) 417 8161
www.stdominiccollege.edu.ph stdominiccollege @sdcapikemen
SCHOOL OF COMMUNICATION, MULTIMEDIA & COMPUTER STUDIES

No.	Module	Test Case	Expected Result	Result
1	Authentication	Register and log in using email/password and Google Sign-In; log in with an unverified email	Verified users are admitted; unverified email/password users are restricted	[Passed]
2	Account approval	Log in as pending, rejected, and disabled users; register a Student and a staff account	Students are auto-approved; unapproved users are blocked from normal use	[Passed]
3	Role-based access	Access admin and super admin routes and APIs using student, faculty, and utility staff accounts	Unauthorized requests are denied	[Passed]
4	Reservation and conflict detection	Submit overlapping reservations and reservations that overlap class schedules	Conflicts are rejected (100% detection)	[Passed]
5	Approval workflow	Submit reservations as Main Campus student, Digital Campus student, and faculty	Correct approval chain is generated per role and campus; invalid approver emails are rejected	[Passed]
6	Document upload	Upload approval documents and open them through signed URLs	Files are stored and accessible only through signed URLs	[Passed]
7	Bluetooth check-in (mobile and web)	Check in near the correct beacon, near a wrong beacon, and with Bluetooth off	Check-in succeeds only when the beacon ID matches the reserved room	[Passed]
8	Presence monitoring	Simulate healthy signal, brief signal loss, out-of-range, and missed heartbeats	Presence status changes among healthy, warning, timed out, and stopped as configured	[Passed]

No.	Module	Test Case	Expected Result	Result
9	Completion and room release	Complete a reservation, then confirm vacancy as utility staff	Monitoring stops on completion; occupancy release is recorded after staff confirmation	[Passed]
10	Notifications	Trigger approval, rejection, and staff alert events	Inbox items are created and push notifications are received	[Passed]
11	Room recommendation	Request a room that is unavailable	Alternative available rooms with similar attributes are suggested	[Passed]
12	Sentiment analysis	Submit positive, neutral, negative, and mixed feedback	Correct sentiment band, aspects, and keywords are stored	[Passed]
13	Gender-disaggregated analytics	View analytics with groups below and above five samples	Groups with fewer than five samples are withheld	[Passed]
14	Bulk schedule import	Upload correctly formatted list and timetable files, files with overlaps, and incorrectly formatted files	Valid rows are imported; invalid and conflicting rows are flagged and skipped	[Passed]
15	DC Space integration	Send valid, duplicate, unauthorized, and unmatched-room requests; change a linked reservation status	Correct 201/200/401/404 responses; status callback is sent	[Passed]
16	Beacon telemetry	Connect and disconnect a device; wait for the interval update	Room beacon fields are updated within the expected interval	[Passed]

Table 4.3 Summary of Unit and System Test Results

4.3.2 Integration Testing

Integration testing verified that data flows correctly across the mobile application, web portal, Next.js backend, Firebase services, Supabase Storage, and ESP32 beacons. All 8 integration scenarios in Table 4.4 [passed / passed after corrections].

No.	Scenario	Components Involved	Expected Result	Result
1	Reservation created on mobile and viewed on web	Mobile, API, Firestore, Web	Same reservation and status appear on both	[Passed]
2	Approval by adviser and building administrator	Web, API, Firestore, Expo push	Status updates and notifications reach the requester	[Passed]
3	Beacon connects to a user device	ESP32, occupancy API, Firestore, Web	Room beacon status updates on the dashboard	[Passed]
4	Bluetooth check-in followed by heartbeats	Mobile, ESP32, API, Firestore	Reservation shows checked-in and presence status	[Passed]
5	Signal loss during an active reservation	Mobile, API, Utility staff app	Warning or timed-out status and staff alert	[Passed]
6	Imported class schedule affects availability	Web, schedule API, Firestore, Mobile	Imported classes block conflicting reservations on both surfaces	[Passed]
7	DC Space event reservation and status callback	DC Space, integration API, Firestore	Linked reservation is created; status changes are returned to DC Space	[Passed]

No.	Scenario	Components Involved	Expected Result	Result
8	Feedback submission to analytics	Mobile/Web, API, VADER, Web	Feedback appears with sentiment and is included in analytics	[Passed]

Table 4.4 Summary of Integration Test Results

4.3.3 Performance Benchmarks

The performance indicators defined in Table 3.3 were measured during pilot testing. Occupancy detection accuracy was computed as the proportion of [n] trials in which the recorded room state matched the actual state. API response time and dashboard update latency were measured over [n] requests, and sentiment analysis accuracy was computed against [n] feedback entries labeled manually by [number] reviewers. Table 4.5 compares the results with the thresholds.

Indicator	Acceptable Threshold	Measured Result	Remarks
Occupancy detection accuracy	≥ 90%	[xx.x%]	[Met]
API response time	≤ 2 seconds	[x.x s]	[Met]
Dashboard update latency	≤ 3 seconds	[x.x s]	[Met]
Conflict detection	100%	[xxx%]	[Met]
BLE detection response time	≤ 5 seconds	[x.x s]	[Met]
Sentiment analysis accuracy	≥ 85%	[xx.x%]	[Met]

Table 4.5 Performance Benchmark Results of e-RoomReserve

[Summarize: state which indicators met the thresholds and explain any that did not. For occupancy detection, note the effect of RSSI variability caused by walls and furniture, as discussed by Janczak et al. (2022).]

4.4 Evaluation Instrument and Treatment of Data

Two questionnaires were used: an Expert Evaluation Questionnaire answered by the 22 IT experts and an End-User Evaluation Questionnaire answered by the 92 organization officers. Each contains 40 statements grouped into eight quality characteristics of ISO/IEC 25010 with five statements each, rated on a five-point Likert scale (1 - Strongly Disagree to 5 - Strongly Agree). The mean of each statement was computed as the sum of the ratings divided by the number of respondents, and the weighted mean of each characteristic was computed as the average of its five statement means. The overall weighted mean is the average of the eight characteristic means. Means were interpreted using the scale in Table 4.6. Because all obtained means fall between 4.00 and 5.00, they are interpreted as either Agree (A) or Strongly Agree (SA).

Mean Range	Verbal Interpretation	Description
4.21 - 5.00	Strongly Agree (SA)	The quality characteristic is fully met
3.41 - 4.20	Agree (A)	The quality characteristic is met
2.61 - 3.40	Neutral (N)	The quality characteristic is moderately met
1.81 - 2.60	Disagree (D)	The quality characteristic is poorly met
1.00 - 1.80	Strongly Disagree (SD)	The quality characteristic is not met

Table 4.6 Likert Scale and Verbal Interpretation

4.5 Results of the Expert Evaluation

This section presents the evaluation of 22 IT experts on the technical quality of e-RoomReserve. Tables 4.7 to 4.14 show the mean rating of each

statement and the weighted mean of each characteristic, and Table 4.15 summarizes the results.

4.5.1 Functional Suitability

Table 4.7 presents the experts' assessment of the functional suitability of e-RoomReserve. The characteristic obtained a weighted mean of [4.xx], interpreted as [Agree/Strongly Agree], which indicates that the experts consider the system to deliver the functions defined in its objectives correctly and completely. Statement [No.] received the highest mean ([4.xx]), while statement [No.] received the lowest ([4.xx]), which still falls within the agree-to-strongly-agree range.

These ratings are consistent with how the functions were implemented. Room occupancy is derived from ESP32 beacons that advertise a room-specific identity, which the mobile application reads and matches against the expected beacon ID before a Bluetooth check-in is accepted. Reservation requests are validated by the backend against class schedules and other reservations before approval, so double bookings are blocked at the server. Reservation status is further governed by the presence-monitoring states (healthy, warning, timed out, and stopped) and by staff confirmation before a room is released, meaning that autonomous control operates together with a human-in-the-loop safeguard. Tying availability to actual presence rather than to scheduled bookings alone follows the vacancy-aware booking approach reported by Somwong et al. (2022), and may account for the favorable ratings on statements 1, 3, and 5.

No.	Statement	Mean	Verbal Interpretation
1	The system accurately detects real-time room occupancy using BLE beacon technology.	[4.xx]	[A/SA]
2	The reservation module correctly processes room booking requests without scheduling conflicts.	[4.xx]	[A/SA]

No.	Statement	Mean	Verbal Interpretation
3	The system's autonomous reservation control functions according to defined business rules (e.g., auto-cancellation for no-shows).	[4.xx]	[A/SA]
4	All integrated modules (web platform, mobile app, and BLE beacons) function correctly without errors.	[4.xx]	[A/SA]
5	The system consistently meets its intended objective of optimizing campus space utilization.	[4.xx]	[A/SA]
Weighted Mean		**[4.xx]**	**[A/SA]**

Table 4.7 Expert Evaluation Results on Functional Suitability

4.5.2 Performance Efficiency

As shown in Table 4.8, the experts rated the performance efficiency of the system with a weighted mean of [4.xx] ([Agree/Strongly Agree]). The highest-rated statement was number [No.] ([4.xx]) and the lowest was number [No.] ([4.xx]). The result suggests that the experts found the system's response times and resource usage acceptable for the pilot deployment.

The ratings reflect the lightweight data path of the system. Beacons post telemetry every 30 seconds and on every connection change, the mobile presence monitor sends heartbeats at the same interval, and the Next.js API routes handle both against Firestore using small payloads. The measured values in Table 4.5 ([API response time] and [dashboard update latency]) corroborate this perception. If the lowest ratings fall on database optimization or high-traffic beacon communication, this would be consistent with the prototype running on free-tier cloud plans and with occupancy history being held in server memory rather than a persistent store (see Section 4.9), both of which limit how far the current setup can be tuned for larger deployments.

No.	Statement	Mean	Verbal Interpretation
1	The system detects and updates room occupancy status within an acceptable response time.	[4.xx]	[A/SA]
2	System resources (server, database, BLE gateway) are utilized efficiently during operation.	[4.xx]	[A/SA]
3	The system maintains stable performance when multiple users reserve rooms concurrently.	[4.xx]	[A/SA]
4	Database queries for room and reservation data are optimized for fast retrieval.	[4.xx]	[A/SA]
5	BLE beacon-to-server communication maintains acceptable performance during high-traffic periods.	[4.xx]	[A/SA]
Weighted Mean		**[4.xx]**	**[A/SA]**

Table 4.8 Expert Evaluation Results on Performance Efficiency

4.5.3 Compatibility

Table 4.9 summarizes the experts' evaluation of compatibility. The weighted mean of [4.xx] ([Agree/Strongly Agree]) shows that the experts agree that the web platform, mobile application, and BLE beacons exchange data effectively and follow accepted integration practices. Statement [No.] was rated highest ([4.xx]) and statement [No.] lowest ([4.xx]).

Compatibility among the three surfaces is achieved by routing all of them through a single backend and data store: the web and mobile clients authenticate through the same Firebase Authentication project and call the same Next.js API routes, while the ESP32 units report to the same occupancy endpoint using a standard BLE GATT service and characteristics for identity. The DC Space integration further demonstrates interoperability with another campus application. It is an authenticated server-to-server exchange in which the DC Space event ID serves as an idempotency key, and reservation status changes are

posted back to the partner system without exposing either database. This approach reflects the concern of Olimat et al. (2023) regarding interoperability and secure data exchange between facility-management platforms.

No.	Statement	Mean	Verbal Interpretation
1	The system exchanges occupancy and reservation data effectively between the web platform, mobile app, and BLE beacons.	[4.xx]	[A/SA]
2	The system adheres to established BLE and API integration standards and protocols.	[4.xx]	[A/SA]
3	Data interoperability between the reservation database and the occupancy detection module is properly supported.	[4.xx]	[A/SA]
4	The system can coexist with other campus applications without conflicts.	[4.xx]	[A/SA]
5	External interfaces (e.g., notifications, authentication) operate consistently and correctly.	[4.xx]	[A/SA]
Weighted Mean		**[4.xx]**	**[A/SA]**

Table 4.9 Expert Evaluation Results on Compatibility

4.5.4 Usability

The experts' evaluation of usability is presented in Table 4.10. The weighted mean of [4.xx] ([Agree/Strongly Agree]) indicates that the experts view the interface as consistent, easy to learn, and efficient for completing reservations. Statement [No.] obtained the highest mean ([4.xx]) and statement [No.] the lowest ([4.xx]).

The interface follows familiar online booking workflows, as anticipated in the operational feasibility analysis (Section 3.3.2). Rooms are browsed by campus, building, and floor; a reservation form collects the required details; an alternative-rooms screen is offered when the preferred room is unavailable; and

role-specific dashboards present only the actions relevant to each user. Status changes are communicated through toast messages, the in-app inbox, and push notifications. Where experts gave lower ratings to error messaging or learning effort, this may point to areas such as the browser requirements of Web Bluetooth check-in (HTTPS and a supported browser), which are not always obvious to first-time users.

No.	Statement	Mean	Verbal Interpretation
1	The user interface follows established usability standards and design principles.	[4.xx]	[A/SA]
2	Navigation elements (room search, booking, dashboard) are logically structured and consistent.	[4.xx]	[A/SA]
3	Error messages provide clear guidance for corrective actions.	[4.xx]	[A/SA]
4	The system minimizes the learning effort required for new users.	[4.xx]	[A/SA]
5	Interface design promotes efficient completion of reservation tasks.	[4.xx]	[A/SA]
Weighted Mean		**[4.xx]**	**[A/SA]**

Table 4.10 Expert Evaluation Results on Usability

4.5.5 Reliability

Table 4.11 shows the experts' ratings on reliability, with a weighted mean of [4.xx] ([Agree/Strongly Agree]). The experts therefore agree that the system operates consistently and can recover from common faults such as beacon disconnection. The highest-rated statement was number [No.] ([4.xx]) while the lowest was number [No.] ([4.xx]).

Reliability is supported by the use of managed cloud services (Firebase Authentication and Firestore) for persistence and by explicit handling of presence

faults. The mobile monitor requires two consecutive beacon warnings before a transient issue is surfaced, the server marks presence as healthy, warning, timed out, or stopped using a two-minute heartbeat window, and staff confirmation serves as the fallback when automated signals are inconclusive. The observed limitations discussed in Section 4.9, namely the absence of an independent server-side scheduler for expiring stale sessions and the non-persistent occupancy history, are consistent with the lower ratings, if any, on recovery, backup, and availability statements.

No.	Statement	Mean	Verbal Interpretation
1	The system operates consistently without unexpected failures.	[4.xx]	[A/SA]
2	The system maintains functionality during prolonged operation and continuous BLE monitoring.	[4.xx]	[A/SA]
3	Recovery mechanisms successfully restore operations after failures (e.g., beacon disconnection).	[4.xx]	[A/SA]
4	Backup and data recovery procedures function correctly for reservation records.	[4.xx]	[A/SA]
5	The system achieves acceptable availability levels for continuous occupancy monitoring.	[4.xx]	[A/SA]
Weighted Mean		**[4.xx]**	**[A/SA]**

Table 4.11 Expert Evaluation Results on Reliability

4.5.6 Security

As presented in Table 4.12, the security of e-RoomReserve obtained a weighted mean of [4.xx] ([Agree/Strongly Agree]) from the experts. Statement [No.] received the highest mean ([4.xx]) and statement [No.] the lowest ([4.xx]).

The result indicates that the experts are satisfied with the authentication, authorization, and data-protection controls of the system.

These ratings correspond to several implemented controls. Access is authenticated through Firebase Authentication with email verification for email/password accounts, and approval status (approved, pending, rejected, or disabled) is stored in Firestore so that unapproved users are blocked from normal use. Roles are normalized and enforced in the protected Next.js API routes rather than only in the client, reservation documents are stored in Supabase Storage and accessed through signed URLs, and the DC Space integration is protected by a server-only Bearer secret. Beacon telemetry is posted to the server over HTTPS. These measures align with the principle of restricting access to authenticated entities and authorized users emphasized by Franco et al. (2025), and with the privacy concerns raised by Li et al. (2023) regarding occupant presence data.

No.	Statement	Mean	Verbal Interpretation
1	User authentication mechanisms effectively prevent unauthorized access to the system.	[4.xx]	[A/SA]
2	The system protects sensitive user and reservation data through appropriate security controls.	[4.xx]	[A/SA]
3	Access permissions are enforced according to user roles (student, faculty, administrator).	[4.xx]	[A/SA]
4	The system records and monitors security-related activities (e.g., login attempts, admin actions).	[4.xx]	[A/SA]
5	Data transmission between BLE beacons, the mobile app, and the server is adequately protected from threats.	[4.xx]	[A/SA]
Weighted Mean		**[4.xx]**	**[A/SA]**

Table 4.12 Expert Evaluation Results on Security

4.5.7 Maintainability

Table 4.13 presents the experts' evaluation of maintainability. The weighted mean of [4.xx] ([Agree/Strongly Agree]) suggests that the experts consider the system easy to modify, diagnose, and maintain. The highest mean was recorded for statement [No.] ([4.xx]) and the lowest for statement [No.] ([4.xx]).

The system is divided into three independent codebases (web and backend, mobile, and hardware firmware), each with clearly separated responsibilities. The web codebase uses TypeScript, Zod for payload validation, shared helper modules for role and campus normalization, and Vitest for automated tests, which makes faults such as booking conflicts or malformed beacon payloads easier to trace. Beacon-specific values (room ID, room name, beacon ID, and Wi-Fi credentials) are configured as constants and secrets rather than embedded in logic. One residual concern is the mixed e-RoomReserve and iRoomReserve naming in package names, URL schemes, and bundle identifiers (Section 4.9), which may explain any lower rating on code organization.

No.	Statement	Mean	Verbal Interpretation
1	The system's architecture supports efficient modification and enhancement.	[4.xx]	[A/SA]
2	The source code is well-organized and maintainable.	[4.xx]	[A/SA]
3	Faults (e.g., beacon errors, booking conflicts) can be identified and corrected with reasonable effort.	[4.xx]	[A/SA]
4	System components (web, mobile, BLE module) are modular and reusable.	[4.xx]	[A/SA]
5	Maintenance activities can be performed with minimal impact on operations.	[4.xx]	[A/SA]
Weighted Mean		**[4.xx]**	**[A/SA]**

Table 4.13 Expert Evaluation Results on Maintainability

4.5.8 Portability

The experts' assessment of portability is shown in Table 4.14. With a weighted mean of [4.xx] ([Agree/Strongly Agree]), the experts agree that the system can be deployed and operated across its supported platforms with limited modification. Statement [No.] was rated highest ([4.xx]) and statement [No.] lowest ([4.xx]).

Portability follows from the technology choices made in Chapter III. The mobile application is built with Expo and React Native and targets both Android and iOS from a single codebase, the web application runs in standard browsers and is deployed on Vercel, and environment-specific settings such as Firebase keys, API base URL, and BLE UUIDs are provided through environment variables. Beacon installation requires flashing the ESP32 firmware and setting room-specific values. Platform dependencies remain in Web Bluetooth, which is available only in supported browsers such as Chrome and Edge, and in background BLE behavior, which differs between Android and iOS.

No.	Statement	Mean	Verbal Interpretation
1	The system can be deployed across multiple platforms with minimal modification.	[4.xx]	[A/SA]
2	Installation and setup procedures (app installation, beacon configuration) are straightforward and efficient.	[4.xx]	[A/SA]
3	The system operates consistently across supported environments (Android, iOS, web browsers).	[4.xx]	[A/SA]
4	Platform-specific dependencies are minimized.	[4.xx]	[A/SA]
5	System migration between environments can be performed efficiently.	[4.xx]	[A/SA]

No.	Statement	Mean	Verbal Interpretation
Weighted Mean		**[4.xx]**	**[A/SA]**

Table 4.14 Expert Evaluation Results on Portability

4.5.9 Summary of the Expert Evaluation

Table 4.15 summarizes the experts' ratings. The overall weighted mean of [4.xx] ([Agree/Strongly Agree]) indicates that the experts consider e-RoomReserve to meet the expected technical quality across all eight characteristics. [Characteristic] obtained the highest weighted mean ([4.xx]), while [characteristic] obtained the lowest ([4.xx]). Even the lowest-rated characteristic remained within the agree range, so no characteristic was rated neutral or below.

Quality Characteristic	Weighted Mean	Verbal Interpretation	Rank
Functional Suitability	[4.xx]	[A/SA]	[#]
Performance Efficiency	[4.xx]	[A/SA]	[#]
Compatibility	[4.xx]	[A/SA]	[#]
Usability	[4.xx]	[A/SA]	[#]
Reliability	[4.xx]	[A/SA]	[#]
Security	[4.xx]	[A/SA]	[#]
Maintainability	[4.xx]	[A/SA]	[#]
Portability	[4.xx]	[A/SA]	[#]
Overall Weighted Mean	**[4.xx]**	**[A/SA]**	

Table 4.15 Summary of Expert Evaluation Results

4.6 Results of the End-User Evaluation

This section presents the evaluation of 92 end-users, all of whom are student organization officers who reserve campus facilities and used the system during the pilot. Tables 4.16 to 4.23 show the ratings for each characteristic, and Table 4.24 summarizes the results.

4.6.1 Functional Suitability

Table 4.16 presents the end-users' assessment of functional suitability. The weighted mean of [4.xx] ([Agree/Strongly Agree]) shows that organization officers found that the system helps them check room availability, reserve rooms without double bookings, and use rooms efficiently. Statement [No.] obtained the highest mean ([4.xx]) and statement [No.] the lowest ([4.xx]).

From the user's perspective, these functions are experienced through the room status screens, the reservation form with conflict checking, and the reservation history and inbox that track approval and check-in. Because the mobile and web applications read from the same backend, a reservation made on one surface appears on the other. The favorable ratings indicate that users saw the automated handling of reservation status, including presence monitoring after Bluetooth check-in and the completion and room-release steps, as a practical improvement over the manual logbook process described in Chapter I.

No.	Statement	Mean	Verbal Interpretation
1	The user can reliably check which rooms are currently occupied or available.	[4.xx]	[A/SA]
2	The user can reserve a room without encountering double-booking issues.	[4.xx]	[A/SA]
3	The system automatically manages my reservation status (e.g., cancels it if I don't check in).	[4.xx]	[A/SA]
4	The web and mobile app work well together when I make or check a reservation.	[4.xx]	[A/SA]

No.	Statement	Mean	Verbal Interpretation
5	The system helps me find and use available rooms efficiently.	[4.xx]	[A/SA]
Weighted Mean		**[4.xx]**	**[A/SA]**

Table 4.16 End-User Evaluation Results on Functional Suitability

4.6.2 Performance Efficiency

As shown in Table 4.17, the end-users rated performance efficiency with a weighted mean of [4.xx] ([Agree/Strongly Agree]). Statement [No.] was rated highest ([4.xx]) and statement [No.] lowest ([4.xx]). Users therefore perceive the application as responsive, with room and schedule information loading promptly.

This perception is supported by the benchmark results in Table 4.5, where the measured API response time and dashboard update latency were [within] the thresholds defined in Chapter III. The 30-second telemetry and heartbeat intervals keep room status reasonably current without overloading the network. Any lower ratings on peak-hour responsiveness would be consistent with the fact that the pilot was run on free-tier services and with a limited number of monitored rooms.

No.	Statement	Mean	Verbal Interpretation
1	Room availability updates quickly when I check the app.	[4.xx]	[A/SA]
2	The app performs smoothly without lag or slowdown.	[4.xx]	[A/SA]
3	The system remains responsive even during peak reservation hours.	[4.xx]	[A/SA]
4	Room and schedule information loads promptly.	[4.xx]	[A/SA]
5	The system performs well even when many rooms are being monitored at once.	[4.xx]	[A/SA]

No.	Statement	Mean	Verbal Interpretation
Weighted Mean		**[4.xx]**	**[A/SA]**

Table 4.17 End-User Evaluation Results on Performance Efficiency

4.6.3 Compatibility

Table 4.18 shows the end-users' ratings on compatibility, with a weighted mean of [4.xx] ([Agree/Strongly Agree]). Statement [No.] received the highest mean ([4.xx]) and statement [No.] the lowest ([4.xx]). Users agree that their reservations sync across web and mobile, that the app works with their devices, and that the room status shown matches what they find on arrival.

The agreement on synchronization results from the shared Firebase Authentication account and the common API and Firestore backend. Agreement that the displayed status matches reality relates to the beacon telemetry that updates each room's connection fields and to the staff confirmation step, which prevents a room from being shown as released before it is physically vacant. Push notifications are delivered through Expo tokens registered by the mobile app, and login is available through email/password or Google Sign-In on both platforms.

No.	Statement	Mean	Verbal Interpretation
1	My reservations sync correctly across the web and mobile app.	[4.xx]	[A/SA]
2	The system works well with the devices I use to access it.	[4.xx]	[A/SA]
3	The room status I see on the app matches what I experience when I arrive at the room.	[4.xx]	[A/SA]
4	Using the system does not interfere with other applications on my device.	[4.xx]	[A/SA]
5	Notifications and login features work as expected.	[4.xx]	[A/SA]

No.	Statement	Mean	Verbal Interpretation
Weighted Mean		**[4.xx]**	**[A/SA]**

Table 4.18 End-User Evaluation Results on Compatibility

4.6.4 Usability

The end-users' evaluation of usability is presented in Table 4.19. The weighted mean of [4.xx] ([Agree/Strongly Agree]) indicates that users found the interface easy to understand and that they learned to use it quickly. Statement [No.] was rated highest ([4.xx]) and statement [No.] lowest ([4.xx]).

This is consistent with the minimal-training expectation stated in the operational feasibility analysis. The reservation flow (choose campus, building, floor, and room; fill in the form; wait for approval; check in near the beacon) follows a familiar online booking pattern, and the alternative-rooms screen helps users recover when their first choice is unavailable. Since the respondents are organization officers who routinely request facilities for meetings, activities, and events, the results also suggest that the approval workflow, which requires an adviser and the building administrator for Main Campus students, was understandable to the users who depend on it most.

No.	Statement	Mean	Verbal Interpretation
1	The system interface is easy to understand.	[4.xx]	[A/SA]
2	It is easy to find and reserve the room I need.	[4.xx]	[A/SA]
3	When errors occur, I understand what I need to do.	[4.xx]	[A/SA]
4	The user learned how to use the system quickly.	[4.xx]	[A/SA]
5	The system helps me complete room reservations easily and efficiently.	[4.xx]	[A/SA]
Weighted Mean		**[4.xx]**	**[A/SA]**

Table 4.19 End-User Evaluation Results on Usability

4.6.5 Reliability

Table 4.20 shows the end-users' ratings on reliability. The weighted mean of [4.xx] ([Agree/Strongly Agree]) suggests that users experienced the system as stable, accessible, and protective of their reservation data. The highest-rated statement was number [No.] ([4.xx]) and the lowest was number [No.] ([4.xx]).

Reservation data is stored in Firestore, a managed cloud database, so records remain intact even when a client session or beacon connection fails. When the Bluetooth signal is lost during an active reservation, the presence monitor and server mark the reservation as warning or timed out instead of discarding it, and staff are prompted to verify the room. Any lower ratings on recovery may relate to the fact that presence-monitor warning push notifications are currently disabled in the mobile service, so users depend on in-app prompts (Section 4.9).

No.	Statement	Mean	Verbal Interpretation
1	The system works reliably whenever I use it.	[4.xx]	[A/SA]
2	The system remains stable during extended use.	[4.xx]	[A/SA]
3	The system quickly recovers when an issue occurs.	[4.xx]	[A/SA]
4	My reservation data is preserved even when problems occur.	[4.xx]	[A/SA]
5	The system is accessible whenever I need it.	[4.xx]	[A/SA]
Weighted Mean		**[4.xx]**	**[A/SA]**

Table 4.20 End-User Evaluation Results on Reliability

4.6.6 Security

As presented in Table 4.21, the end-users rated security with a weighted mean of [4.xx] ([Agree/Strongly Agree]). Statement [No.] obtained the highest mean ([4.xx]) and statement [No.] the lowest ([4.xx]). Users therefore feel that their accounts, personal information, and reservation data are protected and that they only access features appropriate to their roles.

This reflects the role-based dashboards, in which students, faculty professors, utility staff, administrators, and super administrators each see different routes and actions, as well as the approval requirement for non-student accounts. Personal data and uploaded approval documents are not exposed directly, since document access uses signed URLs and API calls are authorized on the server. Users' confidence that activities are tracked corresponds to the notification and status history kept for reservations and rooms.

No.	Statement	Mean	Verbal Interpretation
1	The user feel Unauthorized user cannot access my account.	[4.xx]	[A/SA]
2	My personal information is protected by the system.	[4.xx]	[A/SA]
3	The user can access only the features appropriate to my role.	[4.xx]	[A/SA]
4	The user confident that system activities are properly tracked.	[4.xx]	[A/SA]
5	My personal & reservation data is protected from unauthorized access.	[4.xx]	[A/SA]
Weighted Mean		**[4.xx]**	**[A/SA]**

Table 4.21 End-User Evaluation Results on Security

4.6.7 Maintainability

Table 4.22 presents the end-users' assessment of maintainability as experienced through updates and problem resolution. The weighted mean of [4.xx] ([Agree/Strongly Agree]) indicates that users perceive updates as effective and rarely disruptive. Statement [No.] was rated highest ([4.xx]) and statement [No.] lowest ([4.xx]).

Web updates are released through cloud deployment and reach users without reinstallation, while mobile updates are distributed through Expo. Users can report problems through the inbox and feedback features, which are read by administrators, and the maintenance plan in Section 3.8 provides for regular monitoring, patching, and recalibration. Because the end-user statements concern the effect of changes on daily use rather than code quality, they complement the experts' technical view of maintainability in Table 4.13.

No.	Statement	Mean	Verbal Interpretation
1	Improvements and updates to the system are implemented effectively.	[4.xx]	[A/SA]
2	System updates rarely disrupt my work.	[4.xx]	[A/SA]
3	Problems in the system are resolved promptly.	[4.xx]	[A/SA]
4	System updates & improvements do not negatively affect the features I regularly use.	[4.xx]	[A/SA]
5	The system remains dependable during updates and maintenance.	[4.xx]	[A/SA]
Weighted Mean		**[4.xx]**	**[A/SA]**

Table 4.22 End-User Evaluation Results on Maintainability

4.6.8 Portability

The end-users' evaluation of portability is shown in Table 4.23. The weighted mean of [4.xx] ([Agree/Strongly Agree]) shows that users were able to

access and use the system from different devices without special software or hardware. Statement [No.] received the highest mean ([4.xx]) and statement [No.] the lowest ([4.xx]).

Users can sign in from a browser on a desktop or from the mobile application on Android or iOS with the same account, and their reservations follow them across devices. No dedicated hardware is required on the user's side because the ESP32 beacons are installed in the rooms, and the only device requirement is Bluetooth on the user's phone for check-in. Web Bluetooth check-in remains limited to supported browsers, so mobile check-in is the more consistent path across devices.

No.	Statement	Mean	Verbal Interpretation
1	The user can access the system from different devices (desktop, mobile).	[4.xx]	[A/SA]
2	The system is easy to install or access.	[4.xx]	[A/SA]
3	The system behaves consistently regardless of the device I use.	[4.xx]	[A/SA]
4	The user can use the system without requiring special software or hardware.	[4.xx]	[A/SA]
5	The user can continue using the system smoothly when changing devices or platforms.	[4.xx]	[A/SA]
Weighted Mean		**[4.xx]**	**[A/SA]**

Table 4.23 End-User Evaluation Results on Portability

4.6.9 Summary of the End-User Evaluation

As shown in Table 4.24, the end-users gave an overall weighted mean of [4.xx] ([Agree/Strongly Agree]). [Characteristic] received the highest weighted mean ([4.xx]) and [characteristic] the lowest ([4.xx]). The results show that organization officers accept e-RoomReserve as a dependable, easy-to-use

replacement for the manual reservation process and that the system meets their needs across all eight characteristics.

Quality Characteristic	Weighted Mean	Verbal Interpretation	Rank
Functional Suitability	[4.xx]	[A/SA]	[#]
Performance Efficiency	[4.xx]	[A/SA]	[#]
Compatibility	[4.xx]	[A/SA]	[#]
Usability	[4.xx]	[A/SA]	[#]
Reliability	[4.xx]	[A/SA]	[#]
Security	[4.xx]	[A/SA]	[#]
Maintainability	[4.xx]	[A/SA]	[#]
Portability	[4.xx]	[A/SA]	[#]
Overall Weighted Mean	**[4.xx]**	**[A/SA]**	

Table 4.24 Summary of End-User Evaluation Results

4.7 Comparison of Expert and End-User Evaluations

Table 4.25 compares the weighted means given by the two groups. Both groups rated every characteristic within the agree-to-strongly-agree range, which shows that the technical quality perceived by experts is also experienced by end-users. The largest difference was in [characteristic] ([experts 4.xx vs. end-users 4.xx]). [Interpret the difference. For example, experts may inspect architecture, code organization, and security controls more closely, while end-users mainly judge what they experience through the interface, such as speed, ease of use, and accuracy of room status.] The agreement between the two groups strengthens confidence in the validity of the results, since the experts' technical view is supported by users' day-to-day experience.

Quality Characteristic	Expert Mean	Expert VI	End-User Mean	End-User VI
Functional Suitability	[4.xx]	[A/SA]	[4.xx]	[A/SA]
Performance Efficiency	[4.xx]	[A/SA]	[4.xx]	[A/SA]
Compatibility	[4.xx]	[A/SA]	[4.xx]	[A/SA]
Usability	[4.xx]	[A/SA]	[4.xx]	[A/SA]
Reliability	[4.xx]	[A/SA]	[4.xx]	[A/SA]
Security	[4.xx]	[A/SA]	[4.xx]	[A/SA]
Maintainability	[4.xx]	[A/SA]	[4.xx]	[A/SA]
Portability	[4.xx]	[A/SA]	[4.xx]	[A/SA]
Overall	**[4.xx]**	**[A/SA]**	**[4.xx]**	**[A/SA]**

Table 4.25 Comparison of Expert and End-User Weighted Means

4.8 Achievement of the Objectives

Table 4.26 relates the objectives stated in Chapter I to the evidence gathered in this chapter.

No.	Objective	Evidence	Status
1	Mobile and web application for reservations, bulk scheduling, approval workflows, tracking, and notifications	Sections 4.1.1, 4.1.2, 4.1.4; Table 4.3 (items 1-6, 10, 14)	[Achieved]
2	BLE presence detection that requires proximity for activation, monitors occupancy, and detects vacancy	Section 4.1.3; Table 4.3 (items 7-9, 16); Table 4.5	[Achieved]
3	Automated alerts for signal loss, overtime, and required physical inspection	Section 4.1.4; Table 4.3 (items 8, 10); Table 4.4 (item 5)	[Achieved]

No.	Objective	Evidence	Status
4	Structured reservation workflow that reduces conflicts and unverified use	Table 4.3 (items 3-5); conflict detection result in Table 4.5	[Achieved]
5	Room recommendation, DC Space linking, and gender-disaggregated sentiment analysis	Sections 4.1.5, 4.1.6; Table 4.3 (items 11-13, 15); Table 4.4 (items 7-8)	[Achieved]
6	Analysis of usage logs, occupancy data, and survey responses for actionable insights	Section 4.1.1 (Feedback analytics); Table 4.4 (item 8)	[Achieved]
7	Evaluation of performance, usability, and effectiveness through testing	Tables 4.3 to 4.5; Tables 4.7 to 4.25	[Achieved]

Table 4.26 Achievement of the Objectives of the Study

4.9 Issues and Limitations Observed

Despite the favorable evaluation results, the following limitations were observed in the current implementation and should be considered when interpreting the results:

- Occupancy history received by the occupancy endpoint is stored in server memory and is lost when the server restarts or is redeployed. Only the latest beacon connection fields are persisted in Firestore.
- There is no persistent backend scheduler that independently expires stale presence sessions; the server relies on client heartbeats to detect timeouts.
- Background warning push notifications from the presence monitor are scaffolded but currently disabled; general reservation and administrative push notifications are active.

- Presence detection depends on the reservation holder's phone. It cannot count occupants or detect people who did not reserve a room, and RSSI values can vary with walls, furniture, and interference.
- Web Bluetooth check-in works only over HTTPS or localhost in supported browsers such as Chrome and Edge; mobile check-in is more consistent.
- The internal package names, URL scheme, bundle identifiers, and sample server URL still use the earlier iRoomReserve name.
- The button that opens DC Space from an event reservation in My Reservations is planned but not yet implemented, and failed status callbacks to DC Space are logged but not automatically retried.
- The pilot covered a limited number of rooms, and the end-user sample consisted only of organization officers (92 of 161), so the results may not fully represent other groups such as non-officer students and faculty who do not reserve facilities regularly.

CHAPTER V
SUMMARY OF FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS

This chapter summarizes the development and evaluation of e-RoomReserve, presents the conclusions drawn from the findings in Chapter IV, and offers recommendations for the improvement, deployment, and future study of the system.

5.1 Summary of Findings

The study set out to design and develop an AI-integrated room reservation and occupancy monitoring system that uses Bluetooth Low Energy (BLE) beacons to improve space utilization, enforce reservation policies, and support facility management at St. Dominic College of Asia. The system was developed using the Agile Scrum framework and consists of a Next.js web application that also serves as the backend API, an Expo React Native mobile application, and ESP32 BLE beacon firmware, supported by Firebase Authentication and Firestore, Supabase Storage, and Expo push notifications. It was evaluated by 114 respondents, consisting of 92 organization officers as end-users and 22 IT experts, drawn from a population of 161 organization officers at a 95% confidence level. The findings are as follows:

1. The developed system provides paperless room reservation with server-side conflict checking, a role- and campus-based approval workflow (adviser and building administrator for Main Campus students, building administrator only for faculty and the Digital Campus), document upload with signed-URL access, and reservation tracking through an inbox and push notifications on both web and mobile.

2. BLE presence detection was implemented using ESP32 beacons that advertise a room-specific identity. Users check in by Bluetooth only when the detected beacon matches the reserved room, and the mobile app then sends heartbeats every 30 seconds with a default RSSI threshold of -75. The server classifies presence as healthy, warning, timed out, or stopped, and utility staff physically confirm vacancy before a room is released, which preserves the human-in-the-loop design.

3. The room recommendation module, VADER-based sentiment analysis with gender-disaggregated analytics (withholding groups with fewer than five samples), bulk class-schedule import from Excel files with a validation preview, and an authenticated server-to-server integration with DC Space were all implemented. The DC Space integration went beyond the reference-based link originally proposed by adding an idempotent reservation endpoint and status callbacks, while the two systems still keep separate databases.

4. In system testing, [n] of the 16 unit and system test cases and [n] of the 8 integration scenarios passed. The measured occupancy detection accuracy was [xx.x%], API response time [x.x s], dashboard update latency [x.x s], BLE detection response time [x.x s], conflict detection [xxx%], and sentiment analysis accuracy [xx.x%], which [met / did not meet] the thresholds set in Table 3.3.

5. The 22 IT experts rated the system with an overall weighted mean of [4.xx] ([Agree/Strongly Agree]). Their highest-rated characteristic was [characteristic] ([4.xx]) and their lowest was [characteristic] ([4.xx]); all eight ISO/IEC 25010 characteristics were rated within the agree-to-strongly-agree range.

6. The 92 organization officers rated the system with an overall weighted mean of [4.xx] ([Agree/Strongly Agree]). Their highest-rated characteristic was [characteristic] ([4.xx]) and their lowest was [characteristic] ([4.xx]), and all eight characteristics were likewise rated within the agree-to-strongly-agree range.
7. The two groups agreed closely, with the largest difference between them occurring in [characteristic] ([difference]).
8. Several limitations were observed: occupancy history is held in server memory and lost on restart, there is no independent server-side scheduler that expires stale presence sessions, presence-monitor warning push notifications are disabled, the system cannot count occupants, Web Bluetooth is limited to supported browsers, the earlier iRoomReserve name persists in internal identifiers, and the DC Space shortcut button in My Reservations is not yet implemented.

5.2 Conclusions

Based on the findings of the study, the following conclusions are drawn:

1. e-RoomReserve was successfully developed as a working, integrated platform that replaces the manual logbook and verbal coordination process with a centralized web and mobile reservation system. Its conflict detection, role-based approval, and notification features address the double bookings, miscommunication, and administrative burden described in Chapter I.
2. BLE beacons built on inexpensive ESP32 hardware, at an estimated cost of PHP 470 to PHP 580 per room, are a workable basis for verifying that a reservation holder is physically at the reserved room. Combined with heartbeat-based presence monitoring and staff

confirmation of vacancy, this reduces unverified bookings and no-shows without depending on signal data alone.

3. The room recommendation, sentiment analysis with gender-disaggregated reporting, bulk schedule import, and DC Space integration extend the system beyond basic booking. They give administrators tools to reduce repetitive data entry, avoid conflicts between rooms and events, and base maintenance decisions on user feedback.

4. The system met the performance benchmarks set in Chapter III [except: specify, if any], and passed the unit, system, and integration tests [with noted corrections].

5. The experts and the organization officers who use the system most rated e-RoomReserve in the agree-to-strongly-agree range on all eight quality characteristics. The system is therefore technically sound and acceptable to its intended users, and can be considered ready for extended pilot use.

6. The remaining limitations, which mainly concern persistence, background notifications, and occupant counting, do not prevent the core functions from working but should be resolved before campus-wide production deployment. In its present form the system is best described as a validated prototype rather than a fully hardened production system.

7. All objectives of the study, as summarized in Table 4.26, were [achieved].

5.3 Recommendations

Based on the findings and conclusions, the following recommendations are offered. They are grouped into new features proposed for the next version of the system, technical improvements to the current implementation, institutional and deployment measures, and suggestions for future researchers.

5.3.1 Proposed Feature Enhancements

A. Translation of Feedback

The feedback module currently analyzes free-text comments with VADER, whose lexicon and rules were designed for English. Users of St. Dominic College of Asia often write in Filipino or in Taglish (a mix of Filipino and English), which VADER may score incorrectly or ignore, and this reduces the accuracy of the sentiment bands, detected aspects, and gender-disaggregated results. It is recommended that the feedback pipeline include a language detection and translation step before sentiment analysis. The proposed flow is:

1. Detect the language of each submitted comment on the server.
2. If the comment is not in English, translate it to English using a translation service or a Filipino-capable language model, and run VADER on the translated text.
3. Store the original text together with the translated text and the detected language, so that administrators read the user's own words while analytics use the translated version.
4. Show administrators a toggle between the original and translated comment, and add an optional Filipino-language keyword and aspect list for terms such as "malamig" (cold) or "madumi" (dirty) that translation may miss.

5. Re-measure sentiment accuracy on a labeled set of Filipino, Taglish, and English comments and compare it with the 85% threshold in Table 3.3.

Translation should be treated as an aid and not a replacement for human review, because slang, sarcasm, and code-switching may still be misread. Since feedback text may be sent to an external service, the institution should confirm that this is consistent with its data privacy policy, and should consider a self-hosted or on-device option if it is not.

B. Credit Score System

The system already records the data that a fairness mechanism needs, including check-in time, presence status, completion, and the staff-confirmed release time of every reservation. However, it does not yet use this history to guide user behavior. It is recommended that a credit score system be added to encourage responsible use of rooms, especially among organization officers who reserve frequently. Under the proposed design, each user starts with a fixed base score, which changes according to reservation behavior:

Behavior	Effect on Score	Data Source
Checks in within the grace period and completes the reservation on time	Increase	Check-in and completion timestamps
Cancels early, before a set cut-off time	No change	Cancellation time
Late cancellation	Small decrease	Cancellation time
No-show (reservation released for lack of check-in)	Large decrease	Reservation status
Leaves without marking completion, or occupies the room past the end time	Decrease	Presence status and staff release
Repeated presence time-outs not caused by a beacon fault	Decrease	Presence heartbeat records

Table 5.1 Proposed Credit Score Rules (point values to be set by the administration)

The score can then be used through tiers, for example: users above a high threshold receive priority in approval queues or may reserve further ahead; users in the normal range have standard privileges; and users below a low threshold face shorter advance-booking windows, a limit on active reservations, or a required administrator review. Scores should recover gradually over time and at the start of each semester so that one mistake does not permanently disadvantage a user. To keep the system fair, the following safeguards are recommended:

- Do not penalize a no-show or a time-out when the record shows a beacon fault, such as a disconnected ESP32 unit or a beacon that stopped posting telemetry.
- Provide an appeal process through the existing inbox so that administrators can review and reverse deductions, and record the reason for every manual change.
- Display the score, the recent events that changed it, and the rules to the user, so that the system is transparent.
- Consider the score at the organization level as well as the individual level, since officers reserve on behalf of their organization, and decide with the Student Affairs office who is accountable.
- Treat the score as personal data under the Data Privacy Act of 2012 by limiting who can view it and defining how long it is retained.

C. Smart HVAC and Room Lock

Chapter II noted that e-RoomReserve does not directly control building systems, although the literature identifies occupancy-driven control of building equipment as a major source of energy savings (Chaudhari et al., 2024; Franco et

al., 2025). The room records already contain air conditioning and TV/projector status fields, and the existing ESP32 units already have Wi-Fi and an established connection to the backend. These make a natural next step possible: connecting the room's reservation and presence state to its equipment and door access. It is recommended that a control module be developed with two parts:

- Smart HVAC control. The ESP32 (or a companion controller) switches or signals the room's air conditioner, for example through an infrared blaster or a relay, based on room state. The air conditioner is enabled shortly before an approved reservation begins or upon check-in, and is turned off after the completed reservation is confirmed vacant, or after a configurable period of no presence. Staff receive an alert and a manual override before any automatic shutdown occurs, and the same rule applies to lights and projectors.
- Smart room lock. A smart door lock or electronic strike is unlocked only for the approved reservation window and only after a successful Bluetooth check-in, and is locked again upon completion and staff confirmation. Administrators, utility staff, and security retain master access, and the lock must default to a safe, openable state in case of power or network failure so that no one is trapped and emergency egress is not obstructed.

Because these features act on physical equipment, they should be deployed in stages: first in a monitor-only mode that records what the system would have done, then in a pilot with a small number of rooms, and finally in wider use. Several risks need to be addressed before they are deployed, including the fact that BLE presence reflects only the reservation holder's phone and not the number of people in the room, so equipment should never shut off solely because the holder's phone leaves the range while other occupants remain. A supplementary sensor such as a PIR motion sensor or a people counter is

therefore advised. Fire and safety compliance for door hardware, electrical safety in relay installation, and the facilities office's approval should be settled with the institution. Energy consumption before and after the change should be measured to quantify the savings.

D. Other Recommended Features

- Predictive analytics on utilization. Use the accumulated reservation, no-show, and occupancy data to forecast peak periods, suggest better schedules, and identify rooms that are frequently booked but rarely used.
- Occupant counting. Add low-cost people-counting or motion sensors so the system can detect actual occupancy beyond the reservation holder's phone and identify use of rooms that were never reserved, which is a limitation noted in Chapter I.
- Improved recommendations. Extend the content-based recommendation with historical user preferences and feedback scores per room, so that rooms with poor sentiment on air conditioning, equipment, or internet are ranked lower.
- QR-code check-in as an alternative to Bluetooth. This gives a fallback for devices with Bluetooth problems, for browsers without Web Bluetooth support, and for iOS background limitations.
- Multilingual interface. Offer the application in Filipino as well as English, in line with the feedback translation proposal.
- Organization-level features. Since organization officers are the main users, add organization accounts, shared reservation history, and reminders for the organization's adviser.

- Calendar integration. Allow approved reservations to be exported to Google Calendar or Outlook.

5.3.2 Technical Improvements

- Persist occupancy history in Firestore or another database instead of server memory, so that the data survive restarts and can be used for reports and predictive analytics.
- Add a scheduled backend job (for example, a cloud scheduler or cron function) that expires stale presence sessions and applies automatic no-show release without depending on client heartbeats.
- Enable the presence-monitor warning push notifications that are currently disabled, and test them across Android and iOS background behavior.
- Add automatic retry with backoff for failed status callbacks to DC Space, and implement the button in My Reservations that opens the deployed DC Space application for event reservations.
- Complete the rename from iRoomReserve to e-RoomReserve in package names, the URL scheme, bundle identifiers, and the beacon server URL, taking care to preserve existing installations.
- Signed or authenticated beacon telemetry. Add a per-beacon secret or token so that the occupancy endpoint can reject spoofed posts, and consider rotating BLE identifiers to reduce the risk of replay.
- Conduct load and stress testing with more concurrent users and beacons than in the pilot, and upgrade from free-tier plans before campus-wide deployment.

- Add continuous monitoring, alerting, and a beacon health dashboard, including an alert when a beacon has not posted for a configurable period.
- Increase automated test coverage for the reservation, presence, and integration services, and add a documented backup and restore procedure for Firestore and Supabase data.
- Calibrate the RSSI threshold per room, and repeat the calibration whenever the layout or furniture changes significantly.

5.3.3 Institutional and Deployment Recommendations

- The administration and the ICT Department may adopt e-RoomReserve in phases, starting with rooms that organization officers request most often, and extend it after reviewing the pilot results.
- The Student Affairs office and the ICT Department should agree on reservation policies (grace period for check-in, allowed overtime, penalties, and appeals) before the credit score system is introduced, and publish them to all organizations.
- Organize orientation sessions for organization officers, advisers, utility staff, and building administrators, including guidance on Bluetooth requirements and the check-in procedure.
- Assign responsibility for beacon maintenance, including periodic inspection of USB power adapters and enclosure boxes, replacement of failed units, and firmware updates.
- Establish a data privacy and retention policy for presence data, feedback, and the gender field used in analytics, obtain user consent, and consult the institution's data protection officer.

- Use the sentiment and utilization reports in regular facility planning meetings, so that they lead to actual maintenance and scheduling decisions.

5.3.4 Recommendations for Future Researchers

- Repeat the evaluation with a larger and more varied sample that includes non-officer students, faculty, utility staff, and administrators, and across multiple campuses and semesters.
- Compare room utilization, no-show rates, and administrative time before and after the deployment, in order to measure the actual impact on operations and not only perceived quality.
- Evaluate the accuracy of Filipino and Taglish sentiment analysis, and compare VADER with multilingual or Filipino-trained models.
- Study the fairness, acceptance, and behavioral effects of the credit score system on different groups of users.
- Measure the energy savings and safety outcomes of the smart HVAC and room lock features, along with the accuracy of hybrid occupancy detection that combines BLE with motion or counting sensors.
- Explore other indoor positioning approaches, such as BLE 5.1 direction finding or Ultra-Wideband, for more reliable room-level presence in obstructed environments.

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